

Trade Diversion beyond IIA*

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Abstract

We study how a tariff on one supplier reallocates trade across its competitors. In standard quantitative trade models, competitors gain in proportion to their initial shares, regardless of the trade elasticity. We develop a tractable homothetic demand system that relaxes this restriction. Given expenditure shares and the affected supplier's trade elasticity, the model can generate any positive allocation of the diverted trade, and suppliers' own-price elasticities identify this allocation. We demonstrate that this new model can replicate how U.S. imports of textiles and apparel from China's competitors respond to tariffs on China, capturing the fact that a one-standard-deviation increase in Chinese tariff exposure induces 75 percent greater import growth from Vietnam than from Mexico. Our model and a nested CES alternative both fit this response. Ours requires a trade elasticity for Vietnam above 14, compared with about 8 to 9 in the nested model. Following an increase in Vietnam's price, China absorbs 34 percent of Vietnam's lost share in our model and 81 to 89 percent in the nested alternative.

Keywords. Trade diversion, independence of irrelevant alternatives, trade elasticities

JEL Classification. D11, F11, F13, F14

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1 Introduction

A tariff changes both how much a country imports from the targeted supplier and where it buys the goods that replace those imports. The second adjustment determines which other countries gain access to the market. An estimate of the targeted supplier's trade elasticity helps predict the fall in its sales. To predict how competitors gain trade shares, we need to better capture how buyers substitute suppliers for one another.

Under constant-elasticity-of-substitution (CES) demand, every competitor receives a share of the diverted expenditure proportional to its expenditure share among the unaffected suppliers before the tariff. Independent Fréchet productivity draws in [Eaton and Kortum \(2002\)](#) imply the same conditional sourcing rule. A larger trade elasticity makes imports from the targeted supplier fall more, but does not change how competitors divide the loss. This restriction follows from the independence of irrelevant alternatives (IIA) assumption. Holding competitors' prices fixed, a change in one supplier's price leaves the relative expenditure shares of all other suppliers unchanged.

We show that initial expenditure shares and the targeted supplier's trade elasticity alone do not determine this allocation. Within a globally regular homothetic demand family, we construct models that match these objects and generate any strictly positive reallocation of the supplier's marginal share loss. The construction separates the size of the affected supplier's marginal share loss from the magnitude of that loss, conditional on initial shares and the supplier's trade elasticity. A precise estimate of the first response leaves the second undetermined.

We study this allocation using U.S. textile and apparel imports, where we estimate how growth from China's competitors varies with Chinese tariff exposure across products. Vietnam has a positive response by 2019 and the largest positive response among nine major competitors in 2021. A one-standard-deviation increase in exposure multiplies Vietnam's 2017–2021 import growth factor relative to Mexico's by 1.75, with a 95-percent interval of 1.36 to 2.25. We also estimate supplier differences on one group of products and find that they predict growth in another. Removing Vietnam eliminates this predictive result, while removing Italy does not.

Because relocation and trade rerouting can produce this pattern under CES demand, the estimates do not select a demand system. A CES nest that groups China with Vietnam also fits the Vietnam response. We use this ambiguity to ask what each demand explanation implies for changes in Vietnam's own price and in the prices of several suppliers.

We use the homothetic constant-difference-of-elasticities (HCDE) system of [Hanoch \(1975\)](#).¹ The key feature of HCDE is that supplier-specific substitution parameters affect cross-price responses while initial weights can still match observed expenditure shares. Each supplier has a substitution parameter σ_i . Initial shares determine the demand weights conditional on these parameters, and one equation determines the price index after any finite set of price changes. We use a parameter range in which demand satisfies the usual regularity conditions at every positive price vector.

To see how the construction changes the allocation, let k be the affected supplier and let $q_i = s_i / (1 - s_k)$ be competitor i 's initial share among the other suppliers. For a given trade elasticity $\theta > 0$ of supplier k , we construct a system in which

$$\mathcal{D}_{i \leftarrow k} = q_i r_i, \quad r_i = \frac{\sigma_i - 1}{\theta}, \quad i \neq k. \quad (1)$$

The fraction $\mathcal{D}_{i \leftarrow k}$ measures how much of k 's marginal expenditure-share loss goes to i . The multiplier r_i measures i 's gain relative to proportional diversion. A value above one means that the supplier gains more than CES predicts. We can match any positive allocation d summing to one by setting $r_i = d_i / q_i$, without changing initial shares or k 's trade elasticity.

If we can estimate every supplier's own-price elasticity from price variation and the benchmark shares satisfy the regularity condition $s_i < 1/2$, these responses identify the substitution parameters and the full diversion matrix whenever every initial share is below one half. The implied parameters must exceed one, providing a test of the model. Our application estimates exposure responses across suppliers, leaving their own-price elasticities to be tested with further price variation. Under full pass-through, interpreting the Vietnam–Mexico contrast as demand requires a Vietnam trade elasticity above 14.0 at the aggregate benchmark share. We use this demanding implication to ask what price responses would distinguish the two explanations.

We compare these implications with a China–Vietnam nest fitted to the Vietnam–Mexico contrast. At moderate between-nest elasticities, the nest implies a smaller Vietnam own-price response, but directs much more of Vietnam's lost expenditure back to China. Our normalized example assigns China 34 percent of Vietnam's marginal share loss, compared with 81–89 percent in two nests and 41 percent under proportional diversion. These predictions also extend to simultaneous changes in several prices, as occur in the later tariff episodes.

We choose HCDE because it allows supplier differences with a small number of parameters and remains easy to use for finite counterfactuals. Nested CES requires a partition of suppliers

¹HCDE belongs to the class of homothetic demand systems with indirect implicit additivity characterized by [Matsuyama and Ushchev \(2017\)](#). [Matsuyama \(2023\)](#) uses IIA for indirect implicit additivity. In this paper, IIA denotes independence of irrelevant alternatives. HCDE has the former property while allowing departures from the latter.

and imposes proportional diversion within each nest. The symmetric translog of [Feenstra \(2003\)](#) and [Novy \(2013\)](#) gives equal marginal expenditure-share gains across unaffected varieties. At the country level, these gains depend on the number of varieties. Unrestricted translog and AIDS allow $K(K - 1)/2$ independent symmetric price-interaction coefficients, but global regularity requires additional restrictions. Mixed CES offers another route through a distribution of preferences, possibly related to observed characteristics. Our model uses one substitution parameter per supplier and allows suppliers to be grouped when the data warrant it. This simplicity comes with restrictions across price changes. In particular, the ranking of two rivals does not depend on which other supplier becomes more expensive.

We relate these results to work that uses richer substitution patterns to predict trade reallocation. [Piveteau and Smagghue \(2024\)](#) use product characteristics to predict which countries gain from the trade war, allowing two rivals' ranking to change with the affected exporter. Correlated-productivity models ([Lind and Ramondo, 2023](#)) and nonparametric demand systems ([Adão, Costinot and Donaldson, 2017](#)) also allow patterns that our model rules out. We contribute an explicit construction that holds initial shares and one trade elasticity fixed, an inverse from own-price elasticities to diversion, and predictions for the price responses implied by the Vietnam evidence.

We also show why similar aggregate price responses can conceal different allocations across suppliers. The first derivative of the log price index depends on the affected supplier's initial share, and the second depends on its trade elasticity. Matching these two objects gives agreement through second order, even when competitors gain different fractions of the lost expenditure. Our analysis holds rival prices and nominal expenditure fixed within one import market. General-equilibrium welfare calculations also require equations for supply, prices, and income ([Costinot and Rodríguez-Clare, 2014](#)). [Yang \(2026b\)](#) studies HCDE sourcing in a sectoral trade model. Our counterfactual equations build on [Yang \(2026a\)](#) and the guide to nested trade models in [Kortum and Yang \(2026\)](#).

2 Data

We use annual U.S. Census general-import values from the December files in Schott's public archive, version 2025.4.21 ([Schott, 2008](#)). We aggregate ten-digit codes to eight-digit products in textile and apparel chapters 50–63 for 2015, 2017, 2018, 2019, 2020, and 2021. All reporting origins enter the product-level denominators. Using the [Pierce and Schott \(2012\)](#) concordance and official tariff databases, we select products with stable codes, descriptions, and purely ad valorem general rates at the available dates. The regressions use 1,213 products after excluding HTS 63079098, a broad category that includes masks. We include exporter–product pairs with positive imports at either endpoint, so entry and exit contribute to the estimates.

We measure a product’s exposure to Chinese tariffs by multiplying China’s initial share by the scheduled change in its tariff. A product is more exposed when China initially supplies a larger fraction of imports or faces a larger tariff increase. We use the List 3 and List 4A rates and effective dates to construct calendar-average log gross tariff changes. List 4B was suspended before implementation. For product p , exposure is

$$X_p = s_{Cp,2017} \Delta \ell_{p,2021}, \quad \Delta \ell_{p,t} = \frac{1}{D_t} \sum_{d \in t} \log \frac{1 + \tau_p^0 + a_{pd}}{1 + \tau_p^0}, \quad (2)$$

where D_t is the number of days in year t , τ_p^0 is the general tariff rate, and a_{pd} is the additional duty on imports from China on day d . We use 2021 exposure in each estimation period to compare coefficients in the same units. The 2019 coefficient is therefore not an elasticity with respect to the tariff in place throughout 2019. Online Appendix D.2 describes the product selection and tariff coding.

We use the USITC industry tables for aggregate share calculations and earlier supplier trends (United States International Trade Commission, 2023). These tables cover a different set of products from the regression sample. Calculations that combine product estimates with these aggregate shares are therefore illustrations of the model’s implications. Online Appendix B reports the aggregate allocation and model fits, and Online Appendix D.3 compares within-product reallocation before and after 2017. The regressions below ask whether supplier growth varies systematically with exposure to Chinese tariffs.

3 Evidence on Trade Diversion

3.1 Tariff Exposure and Import Growth

We compare supplier growth across products with different Chinese tariff exposure. Our two-period Poisson pseudo-maximum-likelihood (PPML) specification is

$$\mathbb{E}[V_{ipt} \mid \cdot] = \exp\{\mu_{ip} + \mathbf{1}\{t = b\}[\alpha_p + \gamma_i + \beta_i X_p]\}. \quad (3)$$

We observe imports V_{ipt} from supplier i in product p at an initial date a and final date b . The pair effect μ_{ip} accounts for permanent exporter–product differences. The product effect α_p allows growth to differ across products, and the exporter effect γ_i allows each supplier to grow at a different rate. The coefficient β_i measures whether supplier i grows more in products with greater Chinese tariff exposure, after accounting for these effects. We normalize the nine coefficients to sum to zero and cluster standard errors at 145 four-digit product headings. The normalization leaves fitted values and differences between suppliers unchanged.

Table 1. Import Growth and Chinese Tariff Exposure

Supplier	2015–2017	2017–2019	2017–2021
Vietnam	-0.19 (3.05)	6.10 (2.04)	12.14 (2.70)
India	1.58 (1.45)	-2.99 (2.19)	-3.02 (2.05)
Bangladesh	0.68 (2.45)	2.81 (2.25)	2.27 (4.74)
Mexico	0.45 (2.97)	-2.40 (2.42)	-3.48 (2.41)
Indonesia	2.71 (2.43)	3.44 (3.03)	6.50 (2.76)
Pakistan	0.93 (2.08)	-0.05 (2.49)	-0.91 (3.78)
Cambodia	2.05 (2.68)	-0.17 (4.05)	3.81 (6.26)
Honduras	-8.34 (5.69)	-4.46 (5.60)	-7.75 (5.13)
Italy	0.12 (1.76)	-2.29 (1.97)	-9.57 (2.27)
Vietnam minus Mexico	-0.65 (4.27)	8.50 (3.38)	15.62 (3.55)

Notes. PPML slopes sum to zero across nine suppliers. All periods use 2021 exposure on the same 1,213 products. Standard errors in parentheses cluster by four-digit heading.

Table 1 reports coefficients relative to the nine-supplier mean, with Vietnam’s 2021 estimate equal to 12.14. Relative to Mexico, its coefficient is 15.62 with a standard error of 3.55. Exponentiating this contrast times the standard deviation of exposure gives 1.75, the factor by which expected final-to-initial imports from Vietnam relative to Mexico increase when exposure rises by one standard deviation. The 95-percent interval is 1.36–2.25. Thus the 75-percent increase refers to a ratio of growth factors, rather than a difference in growth rates. It measures a conditional association across products, not a causal tariff effect.

We also find a positive Vietnam–Mexico contrast in 2017–2019, at 8.50 (3.38), before the pandemic and the 2021 Xinjiang order. The 2015–2017 contrast is -0.65 (4.27). To assess whether these responses change over time, we calculate the covariance between estimates from overlapping periods. The 2021 coefficient exceeds the pre-tariff coefficient by 16.27 (5.76), with $p = 0.005$. The two estimates span four tariff-period years and two pre-tariff years. Subtracting twice the pre-tariff coefficient compares the four-year response with an extrapolation of the two-year pre-tariff trend and gives 16.91 (9.49), with $p = 0.077$. The equal-duration test for 2017–2019 and 2015–2017 gives $p = 0.149$. Online Appendix D.5 gives the covariance calculation. Thus the positive tariff-period response is clearer than the evidence of a break from the pre-tariff trend.²

We reject equal responses across the nine suppliers in 2021, with $F_{8,144} = 7.29$ and $p = 4.1 \times 10^{-8}$. The rejection rests on Vietnam and Italy. Removing both and re-estimating gives $p = 0.191$ (Online Appendix D.4). We next ask whether the supplier differences estimated on one set of products predict imports in another.

²Replacing China’s 2017 share in exposure with its 2015 share gives Vietnam–Mexico coefficients of -0.54 (4.56), 10.41 (3.38), and 16.75 (3.64) in the three periods. The positive tariff-period result survives using different base years for exposure and import growth. Initial Chinese specialization can still be correlated with other determinants of growth.

3.2 Out-of-Sample Prediction

We expand the sample to the 25 largest non-China origins by 2017 imports and group the remaining origins together. We randomly divide the four-digit product headings into two groups. For each supplier, we estimate its exposure coefficient using products in the first group. We then multiply that estimate by exposure in the second group and estimate a single coefficient on the resulting variable. A positive coefficient means that suppliers with larger estimated responses in the first group also grow more in exposed products in the second. Product and exporter effects are estimated separately in each group.

We repeat this exercise for 299 fixed divisions of the data and reverse the estimation and prediction groups within each division. Each division contributes the average of its two prediction coefficients. When we exclude a supplier, we remove its observations from both groups without adding them to the pooled remainder.

Table 2 shows that the median prediction coefficient is 0.455 in 2021 and 0.046 in the pre-tariff placebo. The positive tariff-period result survives excluding Italy. Excluding Vietnam reduces the median to 0.144, with $p = 0.410$, and excluding both Vietnam and Italy gives 0.078, with $p = 0.689$. The predictive evidence therefore centers on Vietnam. It motivates a parsimonious model that lets Vietnam’s substitution parameter differ from the other suppliers’, although it does not establish that all other suppliers have identical parameters.

For inference, we combine the 598 one-sided prediction tests using twice their upper median, capped at one, following [Meinshausen, Meier and Bühlmann \(2009\)](#). The rule allows the tests from repeated divisions to be dependent. Its validity requires each conditional test to be valid, which we approximate with a heading-clustered t test. The percentiles in the table measure sensitivity to the division of the data and are not confidence intervals for the median coefficient. The 2017–2019 prediction test is not significant despite Vietnam’s positive individual estimate. Online Appendix D.10 gives the full procedure.

3.3 Economic Interpretation

Relocation and rerouting through Vietnam can generate the estimated responses under CES demand. [Freund et al. \(2024\)](#) show links to Chinese supply chains among countries replacing China, and [Iyoha et al. \(2025\)](#) measure tariff-induced rerouting through Vietnam. Exporter effects remove growth that affects all of a supplier’s products equally, but do not remove supply changes that vary across products. On the demand side, a CES nest that groups China and Vietnam can produce a positive Vietnam coefficient while the other rivals have identical responses. In Section 6, we derive the own-price responses implied by our model and the nest, then show how their predictions differ when several suppliers’ prices change.

Table 2. Out-of-Sample Prediction

Sample	Groups	Median	Split 10–90%	<i>p</i> -value
2017–2021, all groups	26	0.455	[0.287, 0.609]	0.009
Without Italy	25	0.433	[0.260, 0.592]	0.020
Without Vietnam	25	0.144	[-0.053, 0.312]	0.410
Without Vietnam and Italy	24	0.078	[-0.130, 0.261]	0.689
2017–2019, all groups	26	0.171	[-0.044, 0.372]	0.421
2015–2017, all groups	26	0.046	[-0.096, 0.205]	0.851

Notes. We use the same 299 sample splits and 2021 exposure in every row. Products are split by four-digit classification. The range describes variation across splits. The last column combines 598 one-sided tests on products excluded from estimation.

The January 13, 2021 order on Xinjiang cotton can also affect supplier growth because it applies to downstream goods, including goods produced elsewhere with Xinjiang inputs ([U.S. Customs and Border Protection, 2021](#)). Exchange-rate movements change suppliers’ costs as well. The State Bank of Pakistan reports 112.07 rupees per dollar in March 2018 and 133.50 in November, a 16.1-percent fall in the rupee’s dollar value ([State Bank of Pakistan, 2019](#)). Separating substitution in demand from these supply and cost changes requires independent variation in delivered prices.

4 A Model of Trade Diversion

We consider one destination that buys from $K \geq 3$ suppliers. Supplier i charges a delivered price $p_i > 0$ and sells quantity x_i . Total spending is m , and its expenditure share is $s_i = p_i x_i / m$. We assume that all shares are positive and study price changes holding nominal spending and rival prices fixed. Homothetic preferences make expenditure shares independent of the level of spending, so income effects do not account for differences in diversion.

4.1 Proportional Diversion under CES

IIA requires the relative expenditure shares of two suppliers to be independent of the price charged by a third. For distinct i, j, k , the restriction is

$$\mathcal{I}_{ij,k} = \frac{\partial \log(s_i/s_j)}{\partial \log p_k} = 0. \quad (4)$$

CES demand satisfies this condition because $s_i/s_j = (b_i/b_j)(p_i/p_j)^{1-\sigma}$. A change in p_k affects the level of both shares but leaves their ratio unchanged. Since expenditure shares sum to one, all rivals must then gain in proportion to their initial shares. For any finite change in p_k

holding other prices fixed,

$$\mathcal{D}_{i \leftarrow k}^F = \frac{s'_i - s_i}{s_k - s'_k} = \frac{s_i}{1 - s_k} = q_i, \quad s'_k \neq s_k. \quad (5)$$

The numerator is competitor i 's gain and the denominator is supplier k 's loss. Their ratio depends on initial shares. Neither a different elasticity nor a larger price change alters the allocation rule. Canonical Eaton–Kortum sourcing has the same property, with the Fréchet shape parameter replacing $\sigma - 1$. Changes in rival wages or costs can alter the total response in general equilibrium. We study the conditional allocation holding those costs fixed.

4.2 Supplier-Specific Substitution

We allow competitors to respond differently by assigning a separate substitution parameter to each supplier. In the HCDE system, expenditure needed to obtain utility u is $C(p, u) = uP(p)$, where the price index solves

$$\sum_i \beta_i \left(\frac{p_i}{P} \right)^{1-\sigma_i} = 1, \quad \beta_i > 0, \quad \sigma_i > 1. \quad (6)$$

The weights β_i help determine initial shares. The parameters σ_i govern how those shares respond to relative prices. They are not, in general, own-price or pairwise elasticities of substitution. Setting all σ_i equal gives CES.³ Allowing them to differ changes substitution while preserving homotheticity.

We impose regularity globally so that the same demand system can be used for finite price changes. The left-hand side of (6) increases from zero to infinity as P rises, which gives a unique positive price index. Implicit differentiation gives positive expenditure shares. For any price direction z , let $v_i = z_i / p_i$ and $\mu = \sum_i s_i v_i$. We obtain

$$\frac{z^T P_{pp} z}{P} = - \sum_i s_i \sigma_i (v_i - \mu)^2 \leq 0. \quad (7)$$

This expression establishes concavity. Together with smoothness, monotonicity, and homogeneity of degree one, it gives the usual properties of a unit expenditure function at every positive price vector. Online Appendix A.3 derives the expression and gives the corresponding result for non-homothetic CDE.

³Instead, σ_i is a supplier-specific parameter that determines the curvature of the demand system; the implied trade elasticity depends on the equilibrium shares.

4.3 Counterfactual Trade Shares

We calculate counterfactual shares from initial expenditure shares, substitution parameters, and price changes. Conditional on the parameters, initial shares determine the normalized weights

$$\omega_i = \frac{s_i / (\sigma_i - 1)}{\sum_j s_j / (\sigma_j - 1)}. \quad (8)$$

Let $\hat{X} = X' / X$ denote the ratio of the counterfactual to the initial value. The equations for the price index and shares are

$$\sum_i \omega_i \left(\frac{\hat{p}_i}{\hat{P}} \right)^{1-\sigma_i} = 1, \quad s'_i = \frac{s_i (\hat{p}_i / \hat{P})^{1-\sigma_i}}{\sum_j s_j (\hat{p}_j / \hat{P})^{1-\sigma_j}}. \quad (9)$$

We first solve the single equation for $\hat{P}$, whose value lies between the smallest and largest supplier price changes. We then substitute this value into the share equations. Initial shares thus provide the demand weights without further estimation, and suppliers can have the same substitution parameter even if their initial market shares differ.

The price index explains how the system departs from proportional diversion. If only p_k changes, the relative shares of two unaffected suppliers satisfy

$$\frac{s'_i / s'_j}{s_i / s_j} = (\hat{P})^{\sigma_i - \sigma_j}, \quad \mathcal{I}_{ij,k} = s_k (\sigma_i - \sigma_j). \quad (10)$$

A rise in p_k raises the price index. Both rivals become cheaper relative to the import composite, but their responses differ when $\sigma_i \neq \sigma_j$. The supplier with the larger parameter gains relative to the other. If all rivals have the same parameter, their relative shares remain unchanged and diversion is proportional.

5 Trade Diversion and Identification

5.1 The Allocation of Diverted Expenditure

Supplier k 's trade elasticity measures how expenditure on k relative to all other suppliers responds to its delivered price, with a positive sign for a fall in relative expenditure. Local diversion measures each rival's gain as a fraction of k 's marginal expenditure-share loss,

$$\varepsilon_k = -\frac{\partial \log[s_k / (1 - s_k)]}{\partial \log p_k}, \quad \mathcal{D}_{i \leftarrow k} = \frac{\partial s_i / \partial \log p_k}{-\partial s_k / \partial \log p_k}. \quad (11)$$

Under CES, this trade elasticity is $\sigma - 1$, and under canonical Eaton–Kortum sourcing it is the Fréchet shape parameter. Under this definition, suppliers with different initial shares have the same trade elasticity under CES. It differs from the elasticity of quantity demanded and from a tariff coefficient when pass-through is incomplete.

We derive both responses from the same demand equations. With $\bar{\sigma} = \sum_i s_i \sigma_i$,

$$J_{ik} \equiv \frac{\partial \log s_i}{\partial \log p_k} = (1 - \sigma_i) \mathbf{1}\{i = k\} + s_k(\sigma_i + \sigma_k - \bar{\sigma} - 1), \quad (12)$$

$$\mathcal{D}_{i \leftarrow k} = q_i \frac{\sigma_i + \sigma_k - \bar{\sigma} - 1}{\varepsilon_k}. \quad (13)$$

Holding ε_k fixed does not fix the numerator in the second equation. We use this freedom to change the allocation across rivals while preserving the affected supplier's own-price response.

Theorem 1 (Same trade elasticity, arbitrary positive diversion) *Fix interior shares s , an affected source k , and $\theta > 0$. Every strictly positive vector $(d_i)_{i \neq k}$ summing to one is attained by a globally regular HCDE system with those shares and $\varepsilon_k = \theta$. One such system has*

$$\sigma_k = 1 + \theta, \quad \sigma_i = 1 + \theta d_i / q_i \quad (i \neq k). \quad (14)$$

Within the normalization $\sigma_k = 1 + \theta$, the mapping between parameter vectors consistent with $\varepsilon_k = \theta$ and positive diversion vectors is one-to-one. Diversion has $K - 2$ free dimensions.

Proof. All substitution parameters exceed one and their share-weighted mean is $1 + \theta = \sigma_k$. Normalize initial prices and the index to one and set $\beta_i = \omega_i$ in (8). This matches s , and (7) establishes global regularity. Equation (12) gives $J_{kk} = -\theta(1 - s_k)$ and $\partial s_i / \partial \log p_k = s_i s_k(\sigma_i - 1)$, hence $\varepsilon_k = \theta$ and $\mathcal{D}_{i \leftarrow k} = d_i$. Conversely, the normalization and own-price response imply $\bar{\sigma} = \sigma_k$, so (13) uniquely gives (14). $\square$

The construction changes competitors' relative gains while leaving their total gain unchanged to first order. A competitor receives more by having a larger substitution parameter, and the weighted average of these parameters preserves the affected supplier's trade elasticity. We choose the demand weights separately for each model to match the same initial shares, then hold those weights fixed during the price change. Thus the result compares models with the same observable benchmark and the same own-price response. Online Appendix A.8 describes the additional restrictions when some rivals lose share.

5.2 Identification from Own-Price Responses

Because the same parameters determine own-price and cross-price effects, estimates of every supplier's own-price elasticity can identify the parameters and their implied diversion patterns.

Let $a_i = \sigma_i - 1$ and $\bar{a} = \sum_i s_i a_i$. The diagonal of (12) gives

$$(1 - s_i)\varepsilon_i = (1 - 2s_i)a_i + s_i\bar{a}. \quad (15)$$

Each supplier's response depends on its own parameter and the market's share-weighted average. When every supplier accounts for less than half the market, these equations have a unique solution.

Proposition 1 (Identification from trade elasticities) *If every $s_i < 1/2$, trade elasticities measured at the same benchmark uniquely determine all HCDE substitution parameters,*

$$\bar{a} = \frac{\sum_i s_i(1 - s_i)\varepsilon_i / (1 - 2s_i)}{1 + \sum_i s_i^2 / (1 - 2s_i)}, \quad (16)$$

$$a_i = \frac{(1 - s_i)\varepsilon_i - s_i\bar{a}}{1 - 2s_i}. \quad (17)$$

The elasticity vector is consistent with the restriction $\sigma_i > 1$ if and only if every implied $a_i > 0$. In that case it identifies the full diversion matrix and finite counterfactuals.

Proof. Solve (15) for a_i , multiply by s_i , and sum to obtain (16). Equivalently, the coefficient matrix is $\text{diag}(1 - 2s) + ss^\top$, which is positive definite under the share condition. Positive a_i are necessary for $\sigma_i > 1$ and sufficient for regularity by (7). $\square$

The share condition is sufficient, not necessary. Even if all measured trade elasticities are positive, the implied a_i can violate the parameter restriction. This gives a test of the model using own-price responses estimated from price variation. With only two suppliers, the equations have rank one and both trade elasticities equal $s_2 a_1 + s_1 a_2$. The relevant empirical inputs are price derivatives measured at the same benchmark, not elasticities chosen to fit an observed allocation.

The parameters also determine which rivals gain more than CES predicts. Define $\bar{\sigma}_{-k} = \sum_{i \neq k} q_i \sigma_i$. We obtain

$$\frac{\mathcal{D}_{i \leftarrow k}}{q_i} - 1 = \frac{\sigma_i - \bar{\sigma}_{-k}}{\varepsilon_k}. \quad (18)$$

A rival gains more than proportionally exactly when its substitution parameter exceeds the share-weighted average among rivals. Equation (10) gives the corresponding ranking for finite price increases. More generally, the same pair of rivals has the normalized relative response

$$\mathcal{I}_{ij,k} / s_k = \sigma_i - \sigma_j \quad (19)$$

to every other supplier's price. We can test this restriction with price changes affecting two different suppliers. A model in which substitution depends on the similarity between countries can reverse the rivals' ranking and violate the restriction.

5.3 Price Indices and Trade Diversion

An aggregate price index can be well approximated even when the allocation across suppliers is not. To see why, let $f(x)$ be the log change in any smooth homothetic price index when only p_k changes by a factor e^x . Shephard's lemma gives $f'(0) = s_k$. Differentiating again gives $f''(0) = -s_k(1 - s_k)\varepsilon_k$. Thus the initial share determines the first-order price response, and the trade elasticity determines the second-order response. Neither term tells us how competitors divide the lost expenditure.

We compare our construction with CES holding these two objects fixed. Let $V = \sum_i s_i[\sigma_i - (1 + \theta)]^2$. Their log price indices satisfy

$$f_H(x) - f_C(x) = -s_k^3 V x^3 / 6 + O(x^4). \quad (20)$$

The first difference appears at third order, although competitors' expenditure shares can differ at first order. Online Appendix A.5 gives the derivation. When two supplier prices change, cross-price responses affect the index at second order.

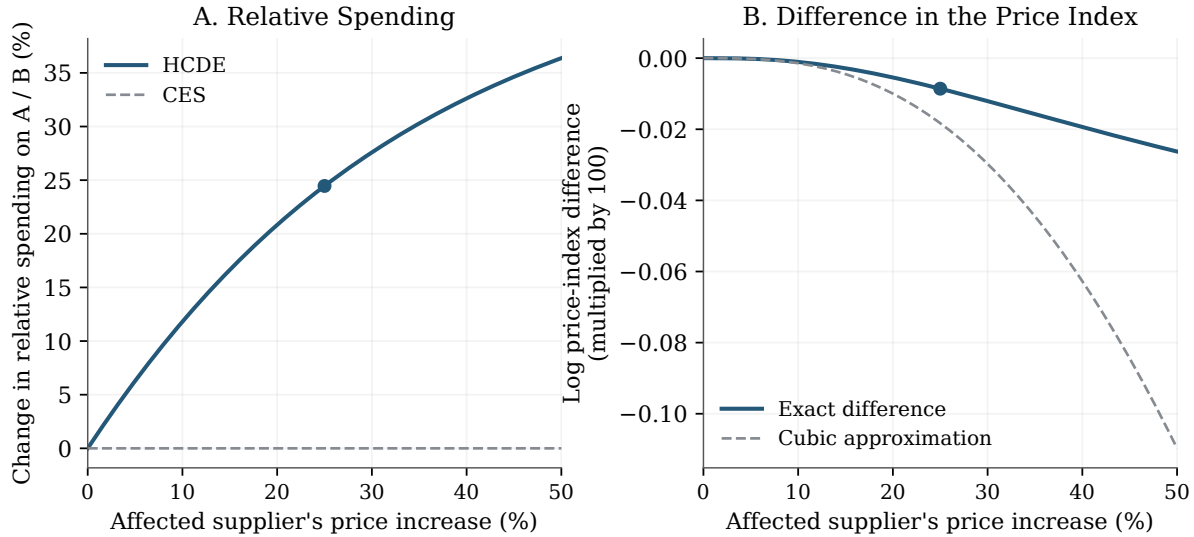


Figure 1. Trade Diversion and the Price Index

Notes. Initial shares are equal. CES has $\sigma = 5$ and our model has $(\sigma_A, \sigma_B, \sigma_k) = (7, 3, 5)$. Only p_k changes. Panel B shows log-index differences multiplied by 100. Markers indicate a 25-percent price increase.

In Figure 1, three suppliers start with equal shares. CES divides the affected supplier's loss equally between its competitors. Our model assigns three-quarters of the marginal loss to one competitor while matching the affected supplier's trade elasticity. With a 25-percent price increase, the competitors' spending ratio changes by 24.46 percent in our model, while the exact difference in log price indices is only -8.59×10^{-5} . Matching the aggregate price response therefore provides little evidence about which competitors gain.

6 Implications for Vietnam

6.1 Vietnam's Own-Price Response

We interpret Vietnam's estimated response to Chinese exposure as a local demand response to ask what it implies for Vietnam's own price. This interpretation requires unchanged rival prices and a pass-through rate ϕ from Chinese tariffs to delivered prices that is the same across products. The coefficient difference then satisfies $\beta_i - \beta_j = \phi(a_i - a_j)$, so full pass-through gives $a_V - a_M = 15.62$ (3.55).

The nine-rival regression identifies parameter differences under these assumptions, leaving their levels and the parameters for China and the remainder unspecified. Positivity nevertheless gives an informative restriction. Since $a_M > 0$, we have $a_V > (\beta_V - \beta_M)/\phi$, while (15) implies $\varepsilon_V > (1 - s_V)a_V$. Therefore,

$$\varepsilon_V > (1 - s_V)(\beta_V - \beta_M)/\phi. \quad (21)$$

With full pass-through and Vietnam's 2018 aggregate share, the right-hand side is 14.03 (3.19). Using Italy instead of Mexico raises the bound to 19.50 (3.30), because the Vietnam–Italy contrast is 21.71 (3.67). The stronger bound uses the largest and smallest of nine estimated slopes, and Italy's large negative coefficient occurs in 2021. We therefore emphasize the Vietnam–Mexico contrast. The standard errors measure uncertainty in the two specified contrasts, without adjustment for selecting the most extreme coefficients or for uncertainty in Vietnam's trade elasticity. Both bounds also require applying product-level parameter differences to an aggregate share. The bounds are informative because they require large own-price responses, while leaving the elasticity unidentified.

Within the 1,213 regression products, Mexico's share falls by 10.54 log percent between 2018 and 2019, against 8.64 for China.⁴ Our model with positive a_i and only China's price changing requires China's share to fall proportionally more than every rival's. Thus no such fit explains all the 2018–2019 share changes in the regression product set. This failure does

⁴The 1,361-product accounting sample gives declines of 9.71 and 7.82 log percent. We calculate both countries' shares using all reporting origins in each product set.

not extend to the full 2017–2021 period, when Mexico’s share falls by 14.16 log percent and China’s by 20.92. Online Appendix C.2 derives the condition and reports the sample checks. The next exercise below is conditional on reading Vietnam’s exposure contrast as demand. Mexico’s earlier decline shows why additional price or supply changes matter for interpreting the episode.

Adding China’s direct tariff response provides further restrictions that can determine parameter levels. In the ten-supplier regression of Online Appendix D.9, some implied levels are negative and violate $a_i > 0$. The lack of identification from the nine-rival contrasts therefore does not imply that the larger regression leaves all levels unrestricted.

Table 3 compares the product-based implications with two aggregate calibrations. The three-moment fit matches the 2019 shares of China, Vietnam, and Bangladesh, while the all-shares fit matches all eleven supplier groups. Both use the 2018 USITC shares and set China’s trade elasticity to four. At the midpoint of the all-shares family, $a_V - a_M = 9.83$. Rescaling the fitted price change to the scheduled tariff change gives 7.56 under full pass-through. The corresponding product-based contrast is 15.62. The product estimates and aggregate calibrations differ in product coverage and identifying assumptions.

For the product-based illustration, we set $a_C = 1.5$, $a_{\text{Italy}} = 0.1$, and the remainder’s parameter equal to Mexico’s. This gives $\varepsilon_V = 20.10$ and $\varepsilon_V/\varepsilon_C = 4.42$, with conditional standard errors of 3.37 and 0.54 from the full coefficient covariance matrix. Varying the unspecified levels gives Vietnam–China ratios of 3.36–4.96 and assigns China 28–35 percent of Vietnam’s marginal loss. Online Appendix D.6 specifies the choices.

With fixed nominal spending, the corresponding quantity elasticity is

$$-\frac{\partial \log x_V}{\partial \log p_V} = 1 + (1 - s_V)\varepsilon_V.$$

The example gives 19.05, well above the own-quantity elasticity of 2.53 in Fajgelbaum et al. (2020). Their estimate pools different products and origins, so it is not a direct test of Vietnam’s response in this market. The large implied response makes an estimate of Vietnam’s own-price elasticity from price variation especially informative.

6.2 A China–Vietnam Nest

A China–Vietnam CES nest explains Vietnam’s response by treating the two countries as especially close substitutes. Let $S = s_C + s_V$, and let η and ω be the between-nest and within-nest elasticities minus one. At the initial shares,

$$\frac{\partial \log(s_V/s_M)}{\partial \log p_C} = \frac{s_C}{S}(\omega - \eta). \quad (22)$$

Table 3. Trade Elasticities Relative to China

Supplier	Aggregate share fits		Product slopes
	Three moments	All shares	Normalized illustration
Vietnam	2.00	2.35–3.57	4.42 (0.54)
Bangladesh	1.63	1.99–3.27	2.57 (0.96)
Mexico	0.72	0.06–1.41	1.37 (0.40)

Notes. Model implications, not own-price estimates. The aggregate fits match the 2019 shares of China, Vietnam, and Bangladesh or all eleven groups, respectively, using 2018 USITC shares and $\varepsilon_C = 4$. The last column fixes $a_C = 1.5$, $a_{\text{Italy}} = 0.1$, and $a_{\text{Other}} = a_M$. Parentheses give conditional delta-method standard errors.

We match the Vietnam–Mexico exposure contrast by setting $\omega - \eta = Sb$ under full pass-through, where $b = 15.62$. Rivals outside the nest have identical relative responses. This matches the local response at the aggregate benchmark. Different nest shares across products would imply different exposure coefficients even with fixed elasticities.

Holding other prices fixed, a rise in Vietnam’s price sends China 41 percent of Vietnam’s marginal share loss under proportional diversion, 34 percent in our model, and 81–89 percent in the two nests in Table 4. The nest directs most of the loss back to China because the two countries are close substitutes within their group. Our model allows Vietnam to gain strongly from China’s loss without directing most of Vietnam’s lost expenditure back to China. We calibrate these models to the Vietnam–Mexico exposure response without imposing a fit to China’s own-price response or every supplier coefficient.

For any positive between-nest parameter, let $q = s_C / (1 - s_V)$. The fitted nest implies

$$\varepsilon_V^N = \eta + qb, \quad \mathcal{D}_{C \leftarrow V}^N = q \frac{\eta + b}{\eta + qb}. \quad (23)$$

At the benchmark, $qb = 6.41$. For $0 < \eta \leq 10$, China receives between 64 percent and a limit of 100 percent of Vietnam’s loss. Its allocation approaches 41 percent as η increases without bound. The nest reaches the Mexico-based bound in our model only at $\eta \geq 7.62$, and the Italy-based bound at $\eta \geq 13.09$. At moderate between-nest elasticities, the nest therefore requires a smaller Vietnam own-price response to explain the same exposure contrast.

6.3 Testing the Models with Several Price Changes

Because the later U.S. tariff episodes affect several suppliers, their implications depend on the full vector of delivered-price changes. Online Appendix D.7 describes the August 2025, February 2026, and July 2026 measures, including rates shared by Vietnam and some of its competitors and the relevant exemptions.

Table 4. Diversion after a Vietnam Price Increase

	China's allocation (%)	Vietnam trade elasticity
Proportional diversion	41.0	—
Our model, normalized illustration	33.9 (2.1)	20.10
China–Vietnam nest, $1 + \eta = 2.5$	88.8 (2.1)	7.91
China–Vietnam nest, $1 + \eta = 4.0$	81.2 (2.9)	9.41

Notes. Local responses at 2018 aggregate shares. Our model and the nests match Vietnam's contrast with Mexico under full pass-through. Parenthesized standard errors condition on each normalization.

To compare the models, let $x_i = \Delta \log p_i$, $z = \Delta \log P$, and z_N be the log change in the China–Vietnam nest price index. If a rival j outside the nest faces the same price change x as Vietnam, the exact relative-share responses are

$$\Delta \log(s_V/s_j) = -(a_V - a_j)(x - z) \quad \text{in our model,} \quad (24)$$

$$\Delta \log(s_V/s_j) = (\omega - \eta)(z_N - x) \quad \text{under nested CES.} \quad (25)$$

The first depends on Vietnam's price relative to the overall index, while the second depends on its price relative to the nest index. Thus the models can give different predictions even when Vietnam and another supplier face equal tariff changes. Equal tariff-point increases imply equal price changes only with the same initial duty and pass-through.

Online Appendix D.7 applies the exact equations to three explicitly specified hypothetical tariff vectors. Vietnam loses 0.15–0.56 log points relative to Bangladesh or Pakistan in our model, while the nests give changes from -0.13 to $+0.26$. These calculations hold the 2018 shares and fitted parameters fixed and do not estimate responses to the later episodes. An empirical test would need product-level duties and exemptions, delivered-price changes, and controls for relocation and rerouting.

7 Conclusion

We construct a globally regular homothetic demand system that matches initial trade shares and the affected supplier's trade elasticity while allowing any positive local allocation of its loss. All suppliers' own-price elasticities identify diversion when each share is below one half, and finite counterfactuals require one price-index equation. We find that Vietnam gains more from exposure to Chinese tariffs than proportional diversion predicts. A China–Vietnam nest can also fit this response. The two demand systems give different predictions when Vietnam's price rises and when several suppliers' prices change.

Online Appendix

A Demand and Trade Diversion

A.1 IIA and Proportional Diversion

We begin by stating the restriction that leads to proportional diversion. Consider a destination that buys from $K \geq 3$ source countries. Let $p_i > 0$ denote the price of goods from source i , $m > 0$ total expenditure, and $s_i = p_i x_i / m$ its expenditure share. All shares are positive. For distinct sources i, j, k , define

$$\mathcal{I}_{ij,k}^C = \frac{\partial \log(s_i/s_j)}{\partial \log p_k} \Big|_u, \quad \mathcal{I}_{ij,k}^M = \frac{\partial \log(s_i/s_j)}{\partial \log p_k} \Big|_m. \quad (\text{A.1})$$

The first derivative holds utility fixed and allows expenditure to adjust. The second holds expenditure fixed and allows utility to adjust.⁵ We refer to the restrictions $\mathcal{I}_{ij,k}^C = 0$ and $\mathcal{I}_{ij,k}^M = 0$ for every distinct triple as compensated and fixed-spending price-IIA. Both say that a change in source k 's price leaves spending on source i relative to source j unchanged. Since p_i and p_j do not change, the restriction also applies to the ratio of their quantities. We hold the set of sources fixed. At least three sources are required to test this restriction.

Under homothetic CES,

$$s_i = \frac{a_i p_i^{1-\sigma}}{\sum_{\ell} a_{\ell} p_{\ell}^{1-\sigma}}, \quad \frac{s_i}{s_j} = \frac{a_i}{a_j} \left(\frac{p_i}{p_j} \right)^{1-\sigma}. \quad (\text{A.2})$$

The price of a third source does not enter this ratio, so both derivatives in (A.1) are zero. We show next that the converse also holds under homotheticity. Requiring price-IIA at every price vector gives CES demand. This result is related to the separability conditions discussed by Matsuyama (2023).

Proposition A.1 (Price-IIA under homotheticity) *Suppose $K \geq 3$ and a homothetic demand system has positive continuously differentiable shares on $\mathbb{R}_{++}^K$ and a concave unit expenditure function. Fixed-spending price-IIA holds for every distinct triple at every price vector if and only if the shares take the CES form in (A.2), with the same $\sigma \geq 0$ for all sources. This includes Cobb–Douglas and the zero-substitution limit.*

Proof. Set $y_i = \log p_i$. Price-IIA makes $\log(s_i/s_j)$ depend only on y_i, y_j . The identity $\log(s_i/s_j) + \log(s_j/s_k) = \log(s_i/s_k)$ then implies $\log(s_i/s_j) = f_i(y_i) - f_j(y_j)$. Homogeneity of shares gives $f'_i(y_i) = f'_j(y_j)$ for arbitrary y_i, y_j , so all derivatives equal a constant ρ .

⁵The same restrictions apply to consumption and production decisions, with output replacing utility in the production case.

Exponentiating and imposing adding up gives $s_i = b_i p_i^\rho / \sum_j b_j p_j^\rho$. Concavity of the corresponding unit expenditure function requires $\rho \leq 1$, so $\sigma = 1 - \rho \geq 0$. The converse follows from the CES share ratios. $\square$

The ratio restriction determines how rivals divide a source's share loss, even for a large price change. For $s'_k \neq s_k$, define

$$\mathcal{D}_{i \leftarrow k}^F = \frac{s'_i - s_i}{s_k - s'_k}. \quad (\text{A.3})$$

Lemma A.1 (Exact proportional diversion) *Under CES, a change in p_k holding other prices fixed implies $\mathcal{D}_{i \leftarrow k}^F = q_i$ for every rival, whenever the denominator is nonzero. The identity does not depend on the elasticity or the size of the price change.*

Proof. Let $Z = 1 - s_k + s_k \hat{p}_k^{1-\sigma}$. Every rival has $s'_i = s_i / Z$. Since the shares sum to one, $s_k - s'_k = (1 - s_k)(1/Z - 1)$. Dividing the rival's gain by this loss gives q_i . $\square$

The same argument applies whenever a change in one source's price leaves all rival share ratios unchanged at fixed expenditure. The finite diversion ratio is undefined if the affected source's share does not change. If its local share response is nonzero, the limit of the diversion ratio as the price change approaches zero is q_i .

In [Eaton and Kortum \(2002\)](#), independent productivity draws $F_i(z) = \exp(-T_i z^{-\theta})$ with the same shape parameter θ imply that destination n buys from source i with share

$$\pi_{ni} = \frac{T_i (c_i \tau_{ni})^{-\theta}}{\sum_\ell T_\ell (c_\ell \tau_{n\ell})^{-\theta}}, \quad \frac{\pi_{ni}}{\pi_{nj}} = \frac{T_i}{T_j} \left(\frac{c_i \tau_{ni}}{c_j \tau_{nj}} \right)^{-\theta}. \quad (\text{A.4})$$

A change in a third source's cost leaves this ratio unchanged when the unit and bilateral costs of i, j are fixed. The resulting trade shares have the CES form with $\theta = \sigma - 1$. This restriction comes from the demand and distributional assumptions. Armington differentiation allows demand systems other than CES, and Fréchet marginal distributions allow correlated productivity draws ([Lind and Ramondo, 2023](#)). In general equilibrium, a shock to k can also change the wages and unit costs of i, j . Their relative trade flows can therefore change even when the conditional demand equation satisfies IIA.

CES elasticities can differ across sectors, as in [Caliendo and Parro \(2015\)](#), and suppliers can be placed in nests. Nested CES permits non-proportional diversion across nests, but requires a choice of which suppliers belong together and preserves IIA within each CES nest. Similarly, CDE preferences over sectoral composites preserve IIA within a CES sourcing nest. Aggregate source ratios can then change because expenditure shifts across sectors or nests, even when every conditional sourcing equation satisfies IIA. We therefore apply the restriction at the level at which sources compete.

A.2 Non-Homothetic CDE

We begin with the CDE expenditure representation

$$\sum_{i=1}^K \beta_i u^{e_i(1-\sigma_i)} \left(\frac{p_i}{C(p, u)} \right)^{1-\sigma_i} = 1, \quad \beta_i > 0, \quad e_i > 0, \quad \sigma_i > 1. \quad (\text{A.5})$$

The expansion parameters e_i govern income effects, while the partial-substitution parameters σ_i govern substitution (Hanoch, 1975). Equal σ_i gives Hanoch's non-homothetic CES (NHCES), as used in models of structural change (Comin, Lashkari and Mestieri, 2021, Sposi, Yi and Zhang, 2026). Equal e_i gives homothetic CDE. We choose the utility index so that this expansion parameter equals one.⁶ We assume $\sigma_i > 1$ to obtain the trade model below. The case $\sigma_i = 1$ requires a separate limiting representation.

Let $t_i = \beta_i u^{e_i(1-\sigma_i)} (p_i/C)^{1-\sigma_i}$. Implicit differentiation gives

$$s_i = \frac{(\sigma_i - 1)t_i}{\sum_{\ell} (\sigma_{\ell} - 1)t_{\ell}}, \quad \bar{\sigma} = \sum_i s_i \sigma_i, \quad \bar{e} = \sum_i s_i e_i. \quad (\text{A.6})$$

The substitution parameters differ from own-price and pairwise elasticities. For $i \neq k$, the Allen–Uzawa elasticity is $\sigma_i + \sigma_k - \bar{\sigma}$ (Surry, 1993, Chen, 2017).

Proposition A.2 (Regularity) *Equation (A.5) defines a unique positive expenditure function at every positive (p, u) . It is smooth, increasing in prices and utility, homogeneous of degree one in prices, and concave in prices.*

Proof. The left-hand side increases strictly from zero to infinity as C increases. Differentiation yields $C_{p_i} = C s_i / p_i > 0$ and $C_u = C \bar{e} / u > 0$. Rescaling prices rescales C . For any direction z , let $v_i = z_i / p_i$ and $\mu = \sum_i s_i v_i$. The price Hessian satisfies

$$\frac{z^T C_{pp} z}{C} = - \sum_i s_i \sigma_i (v_i - \mu)^2 \leq 0. \quad (\text{A.7})$$

Appendix A.3 derives the identity. □

Proposition A.3 (Two forms of IIA) *For distinct i, j, k , CDE implies*

$$\mathcal{I}_{ij,k}^C = s_k (\sigma_i - \sigma_j), \quad (\text{A.8})$$

$$\mathcal{I}_{ij,k}^M = -\frac{s_k}{\bar{e}} [e_i(1 - \sigma_i) - e_j(1 - \sigma_j)]. \quad (\text{A.9})$$

Compensated price-IIA holds for all triples if and only if all σ_i are equal. Fixed-spending price-IIA holds for all triples if and only if all $e_i(1 - \sigma_i)$ are equal.

⁶CES uses the same normalization and sets $\sigma_i = \sigma$ for all sources.

Proof. The log share ratio is a constant plus $[e_i(1 - \sigma_i) - e_j(1 - \sigma_j)] \log u + (1 - \sigma_i) \log p_i - (1 - \sigma_j) \log p_j + (\sigma_i - \sigma_j) \log C$. At fixed utility, $\partial \log C / \partial \log p_k = s_k$. At fixed spending, $C = m$ and $\partial \log u / \partial \log p_k = -s_k / \bar{e}$. Substitution gives both derivatives. Positivity and $K \geq 3$ establish the equivalences. $\square$

Table A1 shows how income and substitution effects affect IIA. Under NHCES, relative demand responds to a third source's price only through income effects, so compensated demand satisfies IIA. HCDE eliminates income effects but allows third-price responses because the substitution parameters differ across sources. In unrestricted CDE, setting $e_i = 1/(\sigma_i - 1)$ makes fixed-spending demand satisfy IIA even when the σ_i differ.

Table A1. Homotheticity and Price-Based IIA

Specification	Homothetic	Compensated IIA	Fixed-spending IIA
CES	Yes	Yes	Yes
NHCES	Not generally	Yes	Equal e_i
HCDE	Yes	Equal σ_i	Equal σ_i
CDE	Not generally	Equal σ_i	Equal $e_i(1 - \sigma_i)$

Notes. Conditions are necessary and sufficient within (A.5), with $K \geq 3$ and rival prices fixed.

A.3 Compensated Derivatives and Concavity

Let $b_i = 1 - \sigma_i < 0$ and $t_i = \beta_i u^{e_i b_i} (p_i / C)^{b_i}$. Differentiating the defining equation gives

$$d \log C = \sum_i s_i d \log p_i + \bar{e} d \log u, \quad s_i = \frac{b_i t_i}{\sum_\ell b_\ell t_\ell}. \quad (\text{A.10})$$

At fixed utility, $d \log t_i = b_i (d \log p_i - d \log C)$. The derivative of the denominator is

$$d \log \left| \sum_\ell b_\ell t_\ell \right| = \sum_\ell s_\ell d \log t_\ell.$$

It follows that

$$\left. \frac{\partial \log s_i}{\partial \log p_k} \right|_u = b_i \mathbf{1}\{i = k\} + s_k (\sigma_i + \sigma_k - \bar{\sigma} - 1). \quad (\text{A.11})$$

For Hicksian demand $h_i = C s_i / p_i$, the elasticity is

$$\frac{\partial \log h_i}{\partial \log p_k} = -\sigma_i \mathbf{1}\{i = k\} + s_k (\sigma_i + \sigma_k - \bar{\sigma}). \quad (\text{A.12})$$

The off-diagonal Allen–Uzawa elasticity is $CC_{ik}/(C_i C_k) = \sigma_i + \sigma_k - \bar{\sigma}$. For $v_i = z_i/p_i$ and $\mu = \sum_i s_i v_i$, the symmetric Hessian therefore satisfies

$$\begin{aligned} \frac{z^\top C_{pp} z}{C} &= - \sum_i s_i \sigma_i v_i^2 + 2 \left(\sum_i s_i \sigma_i v_i \right) \mu - \bar{\sigma} \mu^2 \\ &= - \sum_i s_i \sigma_i (v_i - \mu)^2 \leq 0. \end{aligned} \quad (\text{A.13})$$

For HCDE, shares are independent of utility, so (A.11) also gives the fixed-spending Jacobian. For general CDE, fixed spending imposes $d \log u = - \sum_i s_i d \log p_i / \bar{e}$.

A.4 Diversion Relative to CES

We now relate each rival's gain to its substitution parameter. Given the parameters, we can determine before the shock which rivals gain more than under CES. Let $\bar{\sigma}_{-k} = \sum_{i \neq k} q_i \sigma_i$ denote the share-weighted mean among rivals.

Proposition A.4 (Diversion relative to CES) *For HCDE with $\sigma_i > 1$, local diversion satisfies*

$$\frac{\mathcal{D}_{i \leftarrow k}}{q_i} - 1 = \frac{\sigma_i - \bar{\sigma}_{-k}}{\varepsilon_k}, \quad i \neq k. \quad (\text{A.14})$$

A rival receives more than its proportional allocation if and only if its substitution parameter exceeds $\bar{\sigma}_{-k}$. The ranking also holds for a finite increase in p_k , holding other prices fixed, with $s'_k < s_k$. For any two rivals,

$$\frac{\mathcal{D}_{i \leftarrow k}^F}{q_i} > \frac{\mathcal{D}_{j \leftarrow k}^F}{q_j} \iff \sigma_i > \sigma_j. \quad (\text{A.15})$$

Proof. Equation (12) gives $\varepsilon_k = (1 - s_k)(\sigma_k - 1) + s_k(\bar{\sigma}_{-k} - 1) > 0$. Substituting $\bar{\sigma} = s_k \sigma_k + (1 - s_k) \bar{\sigma}_{-k}$ into (13) gives (A.14). For a finite price increase, $\mathcal{D}_{i \leftarrow k}^F / q_i = (1 - s_k)(s'_i / s_i - 1) / (s_k - s'_k)$. Equation (10) and $\hat{P} > 1$ imply that s'_i / s_i is larger exactly when σ_i is larger. $\square$

In Theorem 1, $\bar{\sigma}_{-k} = 1 + \theta$, so a rival gains more than under CES locally if $r_i > 1$. For a finite shock, the ranking still follows the substitution parameters, but the parameter value that gives an exactly proportional allocation can change with the size of the shock.

The historical calibration determines the parameter differences that match an observed allocation. To use the model for prediction before a shock, we would need to estimate these parameters from independent price variation or relate them to predetermined supplier characteristics.

A.5 Price-Index Differences for Small Shocks

The construction matches the initial shares and the affected source's trade elasticity, so its price index agrees with CES to second order. We derive the first difference between the indices. Let

$s = s_k(0)$, $\sigma_0 = 1 + \theta$, $z_i = \sigma_i - \sigma_0$, and $V = \sum_i s_i(0)z_i^2$. The construction implies $z_k = 0$ and $\sum_i s_i(0)z_i = 0$. Dots denote derivatives with respect to $x = \log \hat{p}_k$, evaluated at zero. For rivals,

$$\dot{s}_i = s_i s(\theta + z_i), \quad \dot{s}_k = -\theta s(1 - s).$$

Since the shares sum to one and the substitution parameters are fixed,

$$\dot{\sigma} = \sum_i \sigma_i \dot{s}_i = \sum_i z_i \dot{s}_i = s \sum_i s_i z_i (\theta + z_i) = sV. \quad (\text{A.16})$$

Along the price path,

$$J_{kk} = 1 - \sigma_k + s_k(2\sigma_k - \bar{\sigma} - 1), \quad \dot{J}_{kk} = \dot{s}_k(2\sigma_k - \bar{\sigma} - 1) - s_k \dot{\sigma}.$$

The benchmark values of s_k , $\dot{s}_k$, and $2\sigma_k - \bar{\sigma} - 1 = \theta$ agree between models. CES has $\dot{\sigma} = 0$, yielding $\dot{J}_{kk,H} - \dot{J}_{kk,C} = -s^2 V$. Because $f' = s_k$ and $\ddot{s}_k = s_k J_{kk}^2 + s_k \dot{J}_{kk}$,

$$f_H'''(0) - f_C'''(0) = -s^3 V.$$

Both log indices are zero initially and have the same first two derivatives. Applying Taylor's theorem gives (20). The first difference between the indices thus depends on the dispersion of the substitution parameters. For the equal-share example, $z = (2, -2, 0)$ and $V = 8/3$, so the cubic coefficient is $-4/243$. The replication compares the exact gap with this approximation at price increases of 1, 5, 10, 25, and 50 percent and checks the limiting coefficient at higher precision.

A.6 Simultaneous Price Changes

When several prices change, diversion can affect the price index to second order. To show this, let $h_i = \log \hat{p}_i$ and $b_i = \sigma_i - (1 + \theta)$ in Theorem 1. The difference between the log-index Hessians is

$$\Delta G = -\text{diag}(s_i b_i) + (s \odot b)s^\top + s(s \odot b)^\top, \quad (\text{A.17})$$

where $\odot$ denotes elementwise multiplication. Thus

$$F_H(h) - F_C(h) = -\frac{1}{2} \sum_i s_i b_i h_i^2 + \left(\sum_i s_i h_i \right) \left(\sum_i s_i b_i h_i \right) + O(\|h\|^3). \quad (\text{A.18})$$

This follows by multiplying each row of (12) by its benchmark share and using $\sum_i s_i b_i = 0$. With changes $h_k = x$ and $h_j = y$, where $j \neq k$ and $b_k = 0$, the second-order term becomes

$$b_j \left[s_j s_k x y - \frac{1}{2} s_j (1 - 2s_j) y^2 \right]. \quad (\text{A.19})$$

The coefficient on xy is $\theta_{sj} s_k (r_j - 1)$. The interaction between the two price changes therefore depends on whether rival j receives more or less diversion than under CES. The y^2 term also differs because the construction does not hold source j 's own elasticity fixed. We illustrate these two effects in the following calculation.

A second source-price change. Using the three-moment calibration in Section B.3, we add a 20 percent increase in Vietnam's source price to the Chinese price change. The composite price index rises by 2.399 percent under HCDE and 2.705 percent under CES. The log-index difference is -29.85 basis points, compared with approximately -0.01 when only China's price changes. Vietnam's final share is 2.77 percent under HCDE and 5.46 percent under CES. Figure A1 shows how the indices and shares change as we vary Vietnam's price.

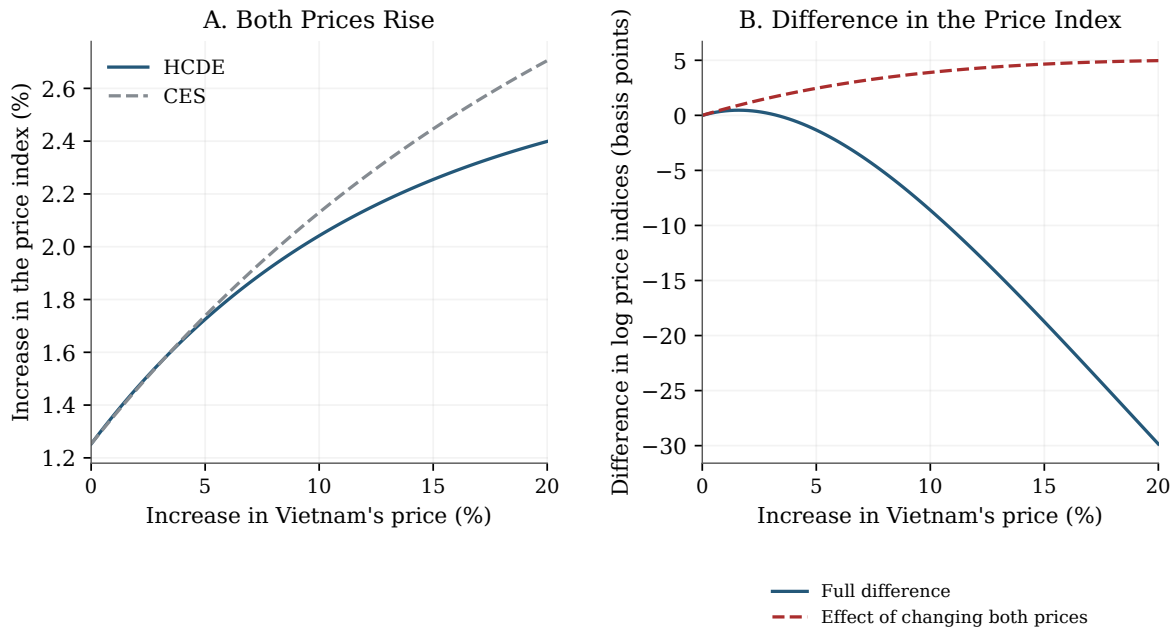


Figure A1. Price Indices When China's and Vietnam's Prices Change

Notes. China's price change and the HCDE substitution parameters use the three-moment fit, and CES has $\sigma = 5$. Panel B reports log-index gaps and mixed interactions in basis points.

We isolate the interaction by subtracting the two single-price responses from the joint response in each model. The difference between these interactions is $+4.97$ log basis points, consistent with the positive cross coefficient in (A.19). The larger negative total gap also reflects the difference in Vietnam's own-price response.

A.7 Restrictions across Shocks

For a fixed rival pair, CDE implies

$$\frac{\mathcal{T}_{ij,k}^C}{s_k} = \sigma_i - \sigma_j, \quad (\sigma_i - \sigma_j) + (\sigma_j - \sigma_\ell) = \sigma_i - \sigma_\ell. \quad (\text{A.20})$$

After division by s_k , the relative response of i and j does not depend on which other source changes price. If we form a matrix using a set of rival pairs and a set of price shifters outside every pair, the matrix has rank at most one. With at least four sources, two independently identified price changes can test this restriction. It does not apply to the full matrix of own- and cross-price responses.

Our model therefore predicts the same ranking of two rivals' relative responses regardless of which other source changes price. Models that relate substitution to supplier characteristics can reverse this ranking (Piveteau and Smagghue, 2024). Our application studies shocks to one exporter and cannot test this difference. Our model also allows some pairs of sources to be compensated complements. Requiring every pair to be compensated substitutes adds $\sigma_i + \sigma_j \geq \bar{\sigma}$, which limits the diversion patterns the model can match.

A.8 Allowing Negative Diversion

A price increase can reduce spending on some rivals even as their combined share rises. We extend the construction to allow these losses while holding the affected source's trade elasticity fixed. Let $a_i = \sigma_i - 1$, $s = s_k$, and $t = \bar{\sigma} - \sigma_k$. For a local diversion vector with $\sum_{i \neq k} d_i = 1$, allowing negative entries, equation (13) and the own-response condition give

$$a_k = \theta - \frac{s}{1-s}t, \quad a_i = \theta \frac{d_i}{q_i} + t \quad (i \neq k). \quad (\text{A.21})$$

All substitution parameters exceed one if and only if

$$-\theta \min_{i \neq k} \frac{d_i}{q_i} < t < \theta \frac{1-s}{s}. \quad (\text{A.22})$$

Substitution into $\bar{\sigma} - \sigma_k = t$ gives the construction and positivity gives the bounds. The interval is nonempty exactly when $d_i > -s_i/s$ for every rival. Theorem 1 sets $t = 0$ and thus selects positive local diversion.

A rival's spending share can fall even when its compensated demand rises. Its Allen–Uzawa elasticity with k is $1 + \theta d_i/q_i$, so it is a compensated complement only when $d_i/q_i < -1/\theta$. Because this larger family still matches the benchmark share and own elasticity, its price index continues to agree with CES to second order.

B Aggregate Trade Reallocation

We use U.S. general imports of textiles and apparel from the general-imports table in [United States International Trade Commission \(2023\)](#). We report ten suppliers separately and group all other origins into a residual category. Total imports rise from 126,992 million in 2018 to 127,238 million in 2019. China’s share falls from 36.864 to 33.632 percent, while Vietnam’s rises from 10.177 to 11.314 percent. These changes relate to the third-country reallocation studied by [Fajgelbaum et al. \(2024\)](#). We use customs-value shares to approximate expenditure shares and calibrate the substitution parameters to the observed allocation, conditional on a change in China’s price holding other prices fixed. Both years come from the same data release. We use earlier editions to examine supplier trends before the tariffs.

B.1 The CES Allocation

Vietnam’s gain divided by China’s loss is 35.17 percent, compared with an initial share of 16.12 percent among non-Chinese suppliers. Lemma [A.1](#) assigns the latter fraction of China’s loss to Vietnam under CES, whatever the elasticity. The implied gain is 0.52 percentage points, compared with 1.14 points in the data. Table [A2](#) reports the observed and implied gains for each supplier.

Table A2. Observed Reallocation and the Exact CES Allocation, 2018–2019

Source	Import share (%)		Change (pp)		Relative allocation
	2018	2019	Observed	CES	
China	36.864	33.632	-3.232	-3.232	—
Vietnam	10.177	11.314	+1.137	+0.521	2.182
India	6.719	6.991	+0.272	+0.344	0.791
Bangladesh	4.475	4.858	+0.384	+0.229	1.675
Mexico	4.702	4.611	-0.091	+0.241	-0.378
Indonesia	3.878	3.839	-0.040	+0.199	-0.200
Pakistan	2.410	2.563	+0.152	+0.123	1.236
Cambodia	1.972	2.225	+0.252	+0.101	2.497
Honduras	2.113	2.299	+0.187	+0.108	1.727
Italy	1.911	1.946	+0.035	+0.098	0.357
Other origins	24.778	25.723	+0.944	+1.268	0.744

Notes. CES allocates China’s observed loss in proportion to rivals’ initial shares. Relative allocation equals one under CES and is negative for a rival share loss. Source. [United States International Trade Commission \(2023\)](#).

Vietnam receives 2.18 times its CES allocation and Cambodia 2.50 times, as measured by $\mathcal{D}_{i \leftarrow C}^F / q_i$. Mexico and Indonesia have negative entries because their shares fall alongside China’s. These ratios describe the observed finite changes. They can differ from the local diversion ratios in Theorem [1](#), which assumes positive diversion to every rival.

We measure the overall departure from CES by summing the absolute differences between observed and CES rival gains and dividing by two. The result is 1.03 percentage points, or 31.8 percent of China’s observed decline. This is the minimum share of total import spending that we must reassign among rivals to move from the conditional CES allocation to the observed shares. The calculation requires no choice of trade elasticity. Figure A2 shows the contribution of each supplier.⁷

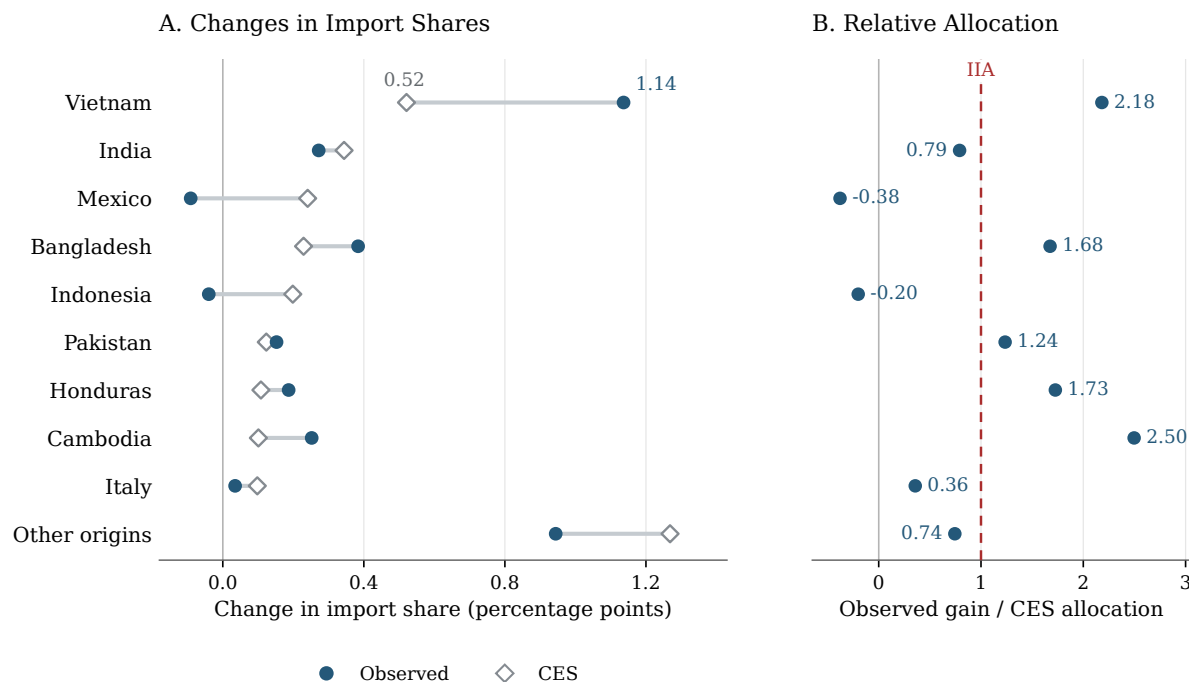


Figure A2. Observed Trade Reallocation and Proportional Diversion, 2018–2019

Notes. Observed and proportional allocations of China's 2018–2019 share loss. Relative allocation equals one under CES. Source: [United States International Trade Commission \(2023\)](#).

B.2 Supplier Trends before the Tariffs

Vietnam gains share well before the tariffs. The 2018 edition of the USITC table reports an annual increase of 0.65 percentage points over 2014–2017, a period when Pakistan and Honduras lose share. Vietnam’s relative allocation over 2014–2018 is 7.80, much larger than its 2018–2019 value. The longer period includes 2018, when tariffs and anticipatory sourcing changes are already under way. We therefore estimate the main trends over periods ending in 2017 and use periods ending in 2018 for sensitivity checks.

⁷Separately identifying all fifteen named suppliers in the USITC table, with the remaining origins in a residual category, raises the discrepancy to 38.2 percent of China’s decline. Grouping the smaller suppliers reduces the measured departure from proportional allocation.

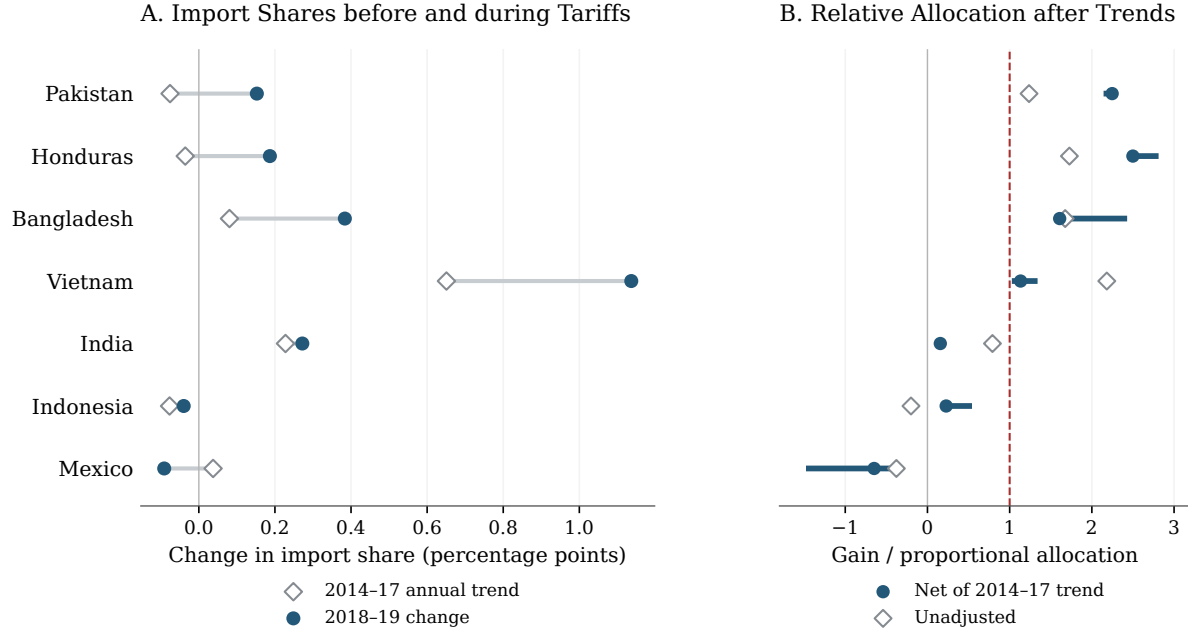


Figure A3. Supplier Trends and Trade Reallocation

Notes. Blue points use 2014–2017 trends. Ranges span the three pre-tariff periods and are not confidence intervals.
Sources. [United States International Trade Commission \(2017, 2019, 2023\)](#).

Let $b_i^{[a,b]}$ be source i 's annual pre-period share change. Define

$$\tilde{\Delta}_i = (s_i^{2019} - s_i^{2018}) - b_i^{[a,b]}, \quad \tilde{R}_i = \frac{\tilde{\Delta}_i}{q_i(-\tilde{\Delta}_C)}, \quad q_i = \frac{s_i^{2018}}{1 - s_C^{2018}}. \quad (\text{B.1})$$

We subtract the trend from China's share change and from each rival's, so that the adjusted changes sum to zero. The denominator is positive for every period we report. A value of one means that the rival's gain beyond trend equals its proportional allocation of China's decline beyond trend. The ratio measures allocation conditional on the estimated trend. The ratio measures allocation conditional on the estimated trend, rather than diversion caused by the tariffs.

After adjustment for trends, Vietnam is close to proportional allocation. Pakistan and Honduras receive more than twice their proportional allocations for all three pre-tariff periods, while Mexico continues to lose share beyond trend. Table A3 and Figure A3 report the adjusted allocations. We also estimate linear trends by least squares and repeat the calculation with periods ending in 2018. The replication reports the departures from proportional allocation under these alternatives.

Table A3. Trade Reallocation Adjusted for Supplier Trends

Source	Unadjusted	Net of own annual trend		
	2018–2019	2012–2017	2014–2017	2015–2017
Vietnam	2.18	1.03	1.13	1.34
India	0.79	0.17	0.16	0.15
Bangladesh	1.68	1.68	1.61	2.43
Mexico	-0.38	-0.39	-0.65	-1.48
Indonesia	-0.20	0.54	0.23	0.30
Pakistan	1.24	2.14	2.25	2.22
Honduras	1.73	2.56	2.50	2.81

Notes. Endpoint annual trends enter (B.1). Growth is calculated within each data vintage. Sources. [United States International Trade Commission \(2017, 2019, 2023\)](#).

B.3 Calibrating Trade Diversion

We use two allocation parameters to connect the theorem to Vietnam’s and Bangladesh’s gains. At $\theta = 4$, let $\eta_i > 0$ be relative weights and define

$$\Lambda = \sum_{\ell \neq C} q_\ell \eta_\ell, \quad r_i = \eta_i / \Lambda, \quad d_i = q_i r_i. \quad (\text{B.2})$$

With all other $\eta_i = 1$, we choose Vietnam’s weight and China’s log price change x to fit

$$s'_C(\eta_V, x) = s_C^{2019}, \quad s'_V(\eta_V, x) = s_V^{2019}. \quad (\text{B.3})$$

The solution gives $\eta_V = 2.72$, $r_V = 2.13$, and a 3.60 percent price increase. Adding Bangladesh’s share as a third moment gives $\eta_V = 3.04$, $\eta_B = 2.36$, and multipliers $r_V = 2.13$, $r_B = 1.66$, and $r_R = 0.70$ for the other rivals.

For the same Chinese price increase of 3.5954 percent, CES and our model give index increases of 1.2526 and 1.2525 percent, a gap of 0.0105 basis points. We assign the same substitution parameter to the other rivals, so the model predicts that their relative shares stay fixed. Their observed share changes can test this restriction.

We can also fit every supplier’s final share. This gives a one-dimensional family of substitution parameters, which we report with the implied trade elasticities in Appendix C.2. Across members of this family, the parameter differences between rivals change by the same factor. We use these relative differences in the product-level test below.

We use the annual tariff schedule to assess the scale of the fitted price change. Weighting the 2018–2019 change in log gross tariffs by China’s 2018 imports gives 4.71 log points. Equating this change to the fitted price change under full pass-through gives $\theta = 3.02$ – 3.13 , with midpoint 3.08. We continue to use $\theta = 4$ in the illustrations. Appendix C.4 explains the rescaling and the implied coefficient of 0.77 in the product-level regression.

B.4 Predicting Later Trade Shares

We next use the 2018–2019 allocation to predict later rival shares, conditional on China’s realized share. We include China, fourteen named rivals, and a residual group. We fit the sixteen initial and final shares, choose the midpoint of the parameter range, and then vary China’s price to match its later share. We use no later rival shares in this calibration.

The model predicts both the ranking of rivals and a linear relationship between their earlier and later growth rates. Let $z_t = \log(P_t/P_{2018})$ and let δ_t denote the corresponding log share denominator. For every rival,

$$\begin{aligned} g_i &\equiv \log(s_i^{2019}/s_i^{2018}) = a_i z_{2019} - \delta_{2019}, \\ \log(s_i^t/s_i^{2019}) &= A_t + B_t g_i, \quad B_t = \frac{z_t}{z_{2019}} - 1, \\ A_t &= B_t \delta_{2019} - (\delta_t - \delta_{2019}). \end{aligned} \tag{B.4}$$

For 2022, the model predicts a slope between 1.91 and 2.15, with midpoint 2.02. Regressing later growth on earlier growth across the fifteen rival groups gives 2.05 (HC3 standard error 0.61). Excluding the residual group gives 1.96 (0.63).

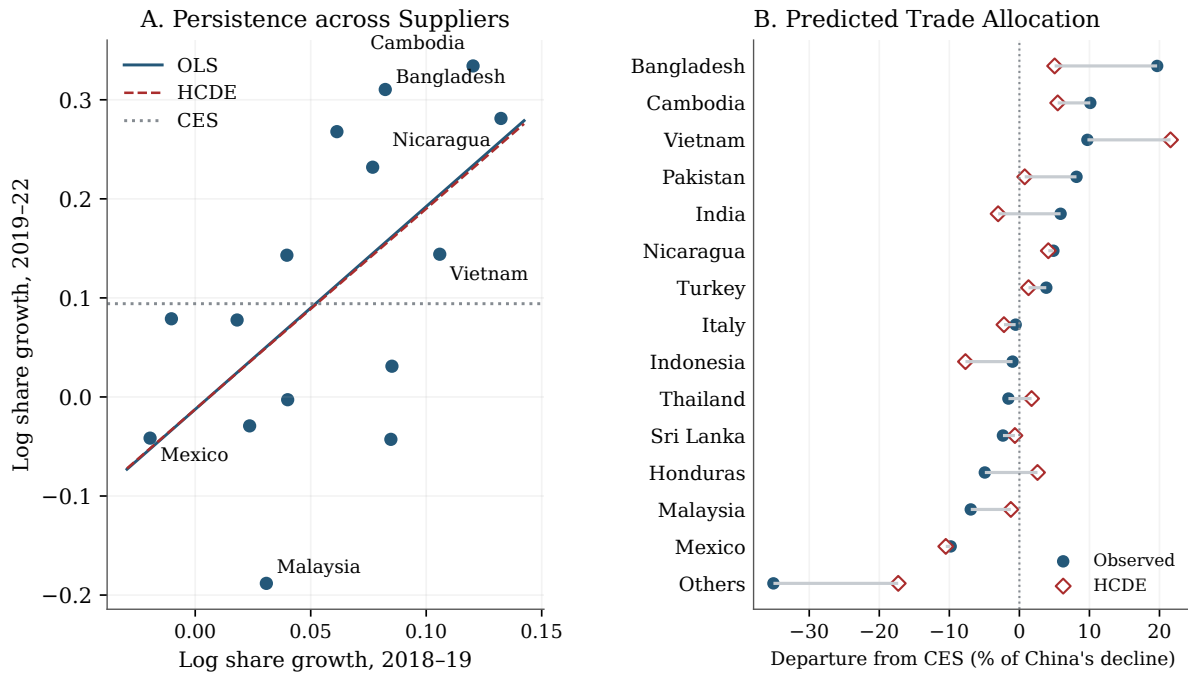


Figure A4. Predicted and Observed Trade Reallocation, 2019–2022

Notes. We calibrate substitution parameters to 2018–2019 shares and choose the price change to match China's 2022 share. We use the midpoint of the calibrated parameter range. Source: [United States International Trade Commission \(2023\)](#).

We measure allocation error as $\frac{1}{2} \sum_{i \neq C} |s_i^t - \hat{s}_i^t| / |s_C^{2019} - s_C^t|$. The numerator is the share of expenditure that must move among rivals to reach the observed allocation. The denominator expresses that amount relative to China's share change. All forecasts condition on the same Chinese share. For the trend forecast, we continue earlier supplier growth and then rescale rival shares to satisfy this condition.

Table A4. Predicted Growth and Allocation Errors

	Observed growth regression			HCDE	Allocation error (%)		
Year	Slope	HC1 SE	HC3 SE	slope	CES	HCDE	Trend
<i>Panel A. USITC industry, original calibration</i>							
2020	0.63	0.90	1.02	-1.76	54.0	81.5	42.3
2021	0.56	1.19	1.31	0.70	210.2	200.3	196.6
2022	2.05	0.56	0.61	2.02	62.2	47.9	52.3
<i>Panel B. Census chapters 50–63, excluding HTS 63079098 throughout</i>							
2020	1.21	0.64	0.90	0.68	146.8	109.9	131.7
2021	1.23	0.61	0.83	1.46	83.4	77.6	83.1

Notes. Errors are normalized by China's absolute share change. Panel B refits all outcomes after excluding HTS 63079098 and has different coverage from Panel A.

In 2022, our model reduces the allocation error from 62.2 percent under CES to 47.9 percent. The full parameter range gives errors of 47.6–48.3 percent, compared with 52.3 percent from continuing supplier growth. We also compare the three forecasts after grouping suppliers into the seven rivals available in every data release and a residual. Our model gives 37.0–38.7 percent, CES 43.4 percent, and pre-tariff supplier trends 35.1–42.0 percent. All three forecasts use the same supplier groups.

China's 2020 share increase is concentrated in mask-related imports ([United States International Trade Commission, 2021](#)). Excluding HTS 63079098 turns its 5.71-point rise in the chapter sample into a 2.00-point decline. We therefore repeat both the calibration and the prediction exercise with this category excluded in every year.

Using the Census sample, our model gives the smallest allocation error in both years, with a 2020 error of 109.9 percent of China's decline, compared with 146.8 for CES and 131.7 for pre-tariff trends. The corresponding 2021 errors are 77.6, 83.4, and 83.1 percent. We cannot reject the predicted slopes of 0.68 and 1.46 against the estimates of 1.21 (0.90) and 1.23 (0.83), with $p = 0.560$ and 0.786 . Excluding the mask-related category thus removes the rejection of the 2020 slope. These forecasts use all chapter products and sixteen supplier groups, so its coverage differs from the stable-product regressions below.

C Calibration to Finite Changes in Trade Shares

C.1 Calibration of Substitution Parameters

We derive the full set of substitution parameters that matches a finite change in shares. Let s and s' be positive share vectors and set $g_i = \log(s'_i/s_i)$. For a price increase at source k , holding other prices fixed, define $x = \log \hat{p}_k > 0$, $z = \log \hat{P}$, and $a_i = \sigma_i - 1 > 0$. The exact-hat equations imply $0 < z < x$ and

$$g_i = a_i z - \delta \quad (i \neq k), \quad g_k = -a_k(x - z) - \delta,$$

where e^δ is the share denominator. Hence $g_i > g_k$ for every rival. The affected source's share must therefore fall proportionally more than every rival's. We show that this condition is also sufficient to fit the observed shares with a globally regular demand system and any initial trade elasticity $\theta > 0$.

To construct the fit, choose

$$\max_{i \neq k}(-g_i) < \delta < -g_k. \quad (\text{C.1})$$

Let $v_i = g_i + \delta > 0$ for rivals and define

$$B(\delta) = \sum_{i \neq k} s_i \frac{e^{v_i} - 1}{v_i}, \quad v_k = \frac{s_k - e^\delta s'_k}{B(\delta)} > 0, \quad \bar{v} = \sum_i s_i v_i. \quad (\text{C.2})$$

We then set

$$z = \frac{(1 - 2s_k)v_k + s_k \bar{v}}{(1 - s_k)\theta}, \quad a_i = \frac{v_i}{z}, \quad x = z \left(1 - \frac{g_k + \delta}{v_k}\right). \quad (\text{C.3})$$

The numerator of z equals $(1 - s_k)^2 v_k + s_k \sum_{i \neq k} s_i v_i > 0$. Thus $z > 0$, every $a_i > 0$, and $x > z$.

For the aggregator, the definition of v_k gives

$$\sum_i \frac{e^\delta s'_i - s_i}{v_i} = 0.$$

Since $a_i = v_i/z$, this is the last equation in (C.6). The share equations hold by construction, so the price index and all final shares are matched. Equation (12) gives

$$\varepsilon_k = \frac{(1 - 2s_k)a_k + s_k \sum_i s_i a_i}{1 - s_k} = \theta$$

by the choice of z . Proposition A.2 establishes global regularity.

Every regular finite fit has a δ in (C.1) and takes this form. The offset satisfies

$$t = \frac{\bar{v} - v_k}{z}, \quad \frac{t}{\theta} = \frac{(1 - s_k)(y - 1)}{1 - 2s_k + s_k y}, \quad y = \frac{\bar{v}}{v_k}. \quad (\text{C.4})$$

As δ increases, each rival's v_i increases and v_k decreases, so y and t increase. Choosing t within its feasible range therefore selects one set of substitution parameters.

We calculate Table A5 as limits at the two boundaries of (C.1). Let $\bar{g}_R = \sum_{i \neq k} s_i g_i / (1 - s_k)$. Adding up and $\varepsilon_k = \theta$ give

$$S_{ik} = 1 + \theta + \frac{g_i - \bar{g}_R}{z}, \quad i \neq k. \quad (\text{C.5})$$

To establish the range of these elasticities, we show that z increases with δ . We have $B > 1 - s_k$ and $0 < B' < B$. Letting $r = e^{g_k + \delta} \in (0, 1)$ gives

$$-v'_k = s_k \left(\frac{r}{B} + \frac{(1-r)B'}{B^2} \right) < \frac{s_k}{1-s_k}, \quad z' = \frac{s_k + (1-s_k)v'_k}{\theta} > 0.$$

Equation (C.5) is therefore monotone in δ for each rival, so its limits at the endpoints give the reported range. At the upper endpoint, $v_k \rightarrow 0$ and $x \rightarrow z(1 + B/s_k)$. At the lower endpoint, at least one rival's substitution parameter approaches one. The strict restriction $\sigma_i > 1$ excludes both endpoints.

C.2 Fitting All Supplier Shares

We next fit every supplier's observed share change. Let $g_i = \log(s'_i/s_i)$, $z = \log \hat{P}$, $x = \log \hat{p}_C$, and δ be the log change in the share denominator.⁸ The exact-hat equations imply

$$a_i = \frac{g_i + \delta}{z} \quad (i \neq C), \quad a_C = -\frac{g_C + \delta}{x - z}, \quad e^\delta \sum_i \frac{s'_i}{a_i} = \sum_i \frac{s_i}{a_i}. \quad (\text{C.6})$$

Our model with positive parameters fits the finite change exactly when $g_C < g_i$ for every rival. The eleven-group USITC shares satisfy this condition, but the stable Census product sets do not for 2018–2019. Within the 1,213 regression products, Mexico's log share change is -0.1054 , compared with -0.0864 for China. The 1,361-product accounting sample gives -0.0971 for Mexico and -0.0782 for China, and the 1,362-product sample specific to that period gives the same values to four decimals. No positive-parameter fit with only China's price changing explains these 2018–2019 aggregate share vectors. The result depends on both the products and the period. For 2017–2021 within the 1,213 regression products, Mexico's log share change is -0.1416 and China's is -0.2092 , so Mexico does not violate this condition. In all Census chapters 50–63, the 2018–2019 changes are -0.0654 for Mexico and -0.0857 for China, and this particular contradiction also does not arise. We use all reporting origins in the denominator for each product set and check the regression-product totals directly against the raw ten-digit records. Additional price or supply changes can alter the implied restrictions.

⁸Changing δ varies the parameters jointly while preserving their ordering and proportional differences among rivals. We restrict δ to keep every $\sigma_i > 1$.

For the USITC fit, setting $\varepsilon_C = 4$ determines the scale, leaving one scalar to select the substitution parameters. Appendix C.1 derives the full family, indexed by

$$\delta_L \equiv \max_{i \neq C}(-g_i) < \delta < \delta_U \equiv -g_C.$$

In the data, $(\delta_L, \delta_U) = (0.019544, 0.091754)$, corresponding to $1.483 < t < 6.851$ for $t = \bar{\sigma} - \sigma_C$. These bounds apply to exact finite changes. The bounds in (A.22) apply to local changes.

Table A5. Substitution Parameters and Elasticities Consistent with the Observed Shares

Source	Parameter σ_i		Allen–Uzawa elasticity S_{iC}	
	$\delta \downarrow \delta_L$	$\delta \uparrow \delta_U$	$\delta \downarrow \delta_L$	$\delta \uparrow \delta_U$
China	4.134	1.000	—	—
Vietnam	11.005	16.214	9.522	9.363
India	5.725	11.119	4.242	4.268
Bangladesh	9.123	14.398	7.640	7.547
Mexico	1.000	6.559	-0.483	-0.292
Indonesia	1.737	7.270	0.254	0.420
Pakistan	7.453	12.786	5.969	5.935
Cambodia	12.155	17.324	10.672	10.473
Honduras	9.317	14.585	7.833	7.734
Italy	4.005	9.459	2.522	2.609
Other origins	5.542	10.943	4.059	4.092
China price change (%)	3.6216	3.7574	—	—
Composite index change (%)	1.2614	1.3074	—	—

Notes. Joint boundary limits of exact 2019 share fits with $\varepsilon_C = 4$. Both endpoints are excluded from $\sigma_i > 1$.

Every parameter vector between the limits in Table A5 fits all eleven final shares. Vietnam’s substitution parameter ranges from 11.005 to 16.214, while its Allen–Uzawa elasticity with China ranges from 9.363 to 9.522. At $\theta = 4$, a rise in China’s price reduces compensated demand for Mexico and raises it for Indonesia for every parameter vector in the range. Both suppliers nevertheless lose expenditure share.

C.3 Implied Trade Elasticities

We use the calibrated parameters to calculate each supplier’s trade elasticity relative to China’s in Table A6. The three-moment fit requires ratios of 2.00 for Vietnam and 1.63 for Bangladesh. Fitting all shares gives ratios of 2.35–3.57 for Vietnam and 0.06–1.41 for Mexico. These restrictions do not depend on the elasticity used to set the scale. Estimates from own-price variation can therefore test them.

Table A6. Implied Trade Elasticities Relative to China

Supplier	Three-moment	All-shares fit		Product slopes
	Ratio	Lower limit	Upper limit	Ratio (s.e.)
China	1.00	1.00	1.00	1.00 (0.00)
Vietnam	2.00	2.35	3.57	4.42 (0.54)
India	0.72	1.18	2.47	1.46 (0.38)
Bangladesh	1.63	1.99	3.27	2.57 (0.96)
Mexico	0.72	0.06	1.41	1.37 (0.40)
Indonesia	0.71	0.22	1.57	3.47 (0.69)
Pakistan	0.71	1.60	2.92	1.91 (0.93)
Cambodia	0.71	2.76	4.03	2.93 (1.47)
Honduras	0.71	2.06	3.36	0.44 (1.35)
Italy	0.71	0.76	2.11	0.05 (0.00)
Other origins	0.80	1.14	2.23	1.40 (0.29)

Notes. Ratios are relative to China. Product-based implications use the normalization in main-text Table 3. Their standard errors condition on that normalization and the benchmark shares. Italy's ratio of 0.05 follows from setting $a_{\text{Italy}} = 0.1$, close to the parameter boundary at zero. It is a consequence of this choice, not an estimated elasticity or a bound on that elasticity.

C.4 Choosing the Trade Elasticity

We show that the fitted allocation does not depend on the chosen trade elasticity. For any $c > 0$, the transformation

$$a_i \mapsto ca_i, \quad \log \hat{p}_i \mapsto c^{-1} \log \hat{p}_i, \quad \log \hat{P} \mapsto c^{-1} \log \hat{P} \quad (\text{C.7})$$

leaves the weights and exponents in (9) unchanged, so it preserves every share while multiplying ε_k by c . The fitted relative weights and diversion multipliers are therefore the same for every $\theta > 0$. The result also holds when we fit all final shares and hold t/θ fixed. Raising θ increases each a_i proportionally and reduces the implied log price change by the same factor. An independently observed price change would help determine this scale.

We use scheduled tariffs to provide such a normalization. Let $\bar{T}$ denote the 2018–2019 change in annual-average log gross tariffs, weighted by China's 2018 imports. We obtain $\bar{T} = 0.04709$. Assuming that pass-through of Chinese tariff changes is constant across products at ϕ , the fitted aggregate price change satisfies $x_{\text{fit}} = \phi \bar{T}$. Since the a_i vary inversely with x_{fit} , the coefficient on the fixed $\theta = 4$ parameter differences is

$$\kappa_{\text{fit}} = \frac{x_{\theta=4}}{\bar{T}} \frac{\sigma_{i,\theta=4} - \sigma_{M,\theta=4}}{d_i} \simeq 0.77, \quad d_i \neq 0. \quad (\text{C.8})$$

The second ratio expresses each fitted parameter vector relative to the fixed direction used in the regressions. It is the same for every rival, and pass-through cancels in this calibration. At full pass-through, $\theta = 4x_{\theta=4}/\bar{T} = 3.02\text{--}3.13$, with midpoint 3.08. The three-moment fit

gives $\theta = 3.00$. We combine scheduled tariffs for the stable chapter sample with an aggregate price change from the USITC industry series. Differences in coverage and between scheduled and paid duties can make these two price measures differ. The rescaling in (C.7) remains exact within the model. The evidence of complete pass-through in [Fajgelbaum et al. \(2020\)](#) motivates this normalization. We must also account for the difference between customs values and duty-inclusive expenditure, as we do for China’s own coefficient in Section D.9.

D Data and Estimation

D.1 Aggregate Data

We take the aggregate series from the numerical entries in the public HTML general-imports table of [United States International Trade Commission \(2023\)](#). USITC extracts these data from DataWeb/Census on February 16, 2023. The table is numbered TX.2 in HTML and TX.11 in the companion PDF. In the eleven-group calibration, we identify China, Vietnam, India, Bangladesh, Mexico, Indonesia, Pakistan, Cambodia, Honduras, and Italy separately. Other origins equal total imports less these ten sources. The prediction exercise with fifteen rivals includes every named supplier and the residual.

We obtain earlier shares from Table TX.1 of the 2016 and 2018 editions. Both report Vietnam, India, Bangladesh, Mexico, Indonesia, Pakistan, and Honduras separately, and we recompute the residual using these groups. We calculate growth within each data release. For 2012–2017, we combine the 2012–2014 change from the 2016 edition with the 2014–2017 change from the 2018 edition. The 2014–2017 and 2015–2017 trends each use one release. Our main trends divide the endpoint share change by the number of years. We check sensitivity using least-squares trends and periods ending in 2018. Both 2018 and 2019 outcomes come from the 2022 edition.

D.2 Product Data and Scheduled Tariffs

We aggregate year-to-date general-import values from the December Census files in Schott’s version 2025.4.21 archive. The files cover 2015, 2017, 2018, 2019, 2020, and 2021. We sum ten-digit codes to eight digits in chapters 50–63 and include all reporting origins when calculating each product’s total imports. To construct a stable 2015–2019 sample, we exclude any code linked to a different eight-digit code in the 2015–January 2019 concordance of [Pierce and Schott \(2012\)](#). We require positive world imports in all four years and purely ad valorem general tariff rates that do not change across the official 2015, 2017, and 2019 databases. These restrictions leave 1,367 products, covering 80.7 percent of chapter imports in 2017 and 80.1 percent in 2019.

For the post-tariff accounting, we exclude five products supplied entirely by China in 2018, leaving 1,362 products. Requiring China’s share to be below one in 2015 as well gives the 1,361 products used in both periods of Table A7. For the tariff regressions, we require an interior Chinese share in 2017. This leaves 1,270 products and 11,430 possible exporter-product pairs

for the nine rivals. Of these, 4,804 have positive flows in both 2017 and 2019, 872 enter, 657 exit, and 5,097 have no flow at either endpoint. Requiring positive flows in 2015 leaves 4,414 pairs. We then remove products with only one remaining rival, giving 4,224 pairs and 1,007 products for OLS. PPML includes pairs with positive trade at either endpoint and products with positive total imports from rivals in both years. This gives 6,285 pairs, including 844 entrants, 637 exits, and continuing pairs excluded by the OLS restriction on 2015 trade.

The tariff assignment has 917 List 3, 652 List 4A, and 111 suspended List 4B chapter codes. List 3 adds 10 percent from September 24, 2018 and 25 percent from May 10, 2019. List 4A adds 15 percent from September 1, 2019. List 4B is not implemented. For unchanged general rate τ_p^0 and scheduled additional duty a_{pd} on day d , we use

$$\Delta \ell_{p,t} = \frac{1}{D_t} \sum_{d \in t} \left[\log(1 + \tau_p^0 + a_{pd}) - \log(1 + \tau_p^0) \right]. \quad (\text{D.1})$$

The three tariff lists divide products into three policy groups. Most variation in exposure comes from China's initial share. In the main product sample, 78.3 percent of exposure variance lies within these groups, including variation in general tariff rates. Mean exposure across exporter-product pairs is 0.0271, with standard deviation 0.0282. Across the 1,007 distinct products, the corresponding values are 0.0299 and 0.0307. The replication files report these moments separately for each exporter in the 2019 OLS sample.

For the 2021 OLS long difference, we also require product descriptions and purely ad valorem general tariff rates to remain unchanged between 2019 and the June 2021 database. Trade must be positive in 2015, 2017, and 2021, but can be zero in 2019. The 2021 exposure uses a full year at 25 percent for List 3 and 7.5 percent for List 4A. The latter rate takes effect on February 14, 2020 ([U.S. Customs and Border Protection, 2020](#)). Exposure measures the annual tariff level at the endpoint relative to 2017. It does not sum tariffs over the intervening years. The sample has 4,202 pairs and 1,005 products. We adjust for trends by subtracting twice the 2015–2017 log growth from the four-year change. The annual PPML estimates include entry and exit for the same 1,214 products in every year, or 1,213 when we exclude HTS 63079098. The 2021 PPML samples have 6,419 and 6,410 pairs, respectively. The concordance checks product descriptions and general tariff rates at the available dates.

D.3 Within-Product Proportional Allocation

We use annual U.S. general-import values in HTS chapters 50–63 from the Census files distributed by [Schott \(2008\)](#). The accounting sample has 1,361 stable eight-digit products with unchanged ad valorem general tariff rates and covers about 80 percent of chapter imports. We require China's share to be below one at both initial dates, 2015 and 2018, so that we compare the same products in both periods. This sample differs from the USITC industry aggregate. Appendix [D.2](#) describes its construction.

Let w_{pt} be product p 's share of sample imports and s_{ipt} exporter i 's share within that product. Then $S_{it} = \sum_p w_{pt} s_{ipt}$. Conditional on each product's Chinese share at the end of period a – b , the CES allocation is

$$\tilde{s}_{ip,b} = s_{ip,a} \frac{1 - s_{Cp,b}}{1 - s_{Cp,a}}, \quad i \neq C. \quad (\text{D.2})$$

Set $\tilde{s}_{Cp,b} = s_{Cp,b}$. The exact decomposition is

$$\begin{aligned} \Delta S_i &= C_i + P_i + R_i, \\ C_i &= \sum_p (w_{p,b} - w_{p,a}) s_{ip,a}, \\ P_i &= \sum_p w_{p,b} (\tilde{s}_{ip,b} - s_{ip,a}), \quad R_i = \sum_p w_{p,b} (s_{ip,b} - \tilde{s}_{ip,b}). \end{aligned} \quad (\text{D.3})$$

The decomposition separates three sources of an exporter's aggregate share change. The first term, C_i , measures shifts in spending across products. The second, P_i , is the proportional allocation of China's share change within products. The third, R_i , measures departures from this allocation. We include entry and exit. If an exporter initially sells none of a product, its proportional allocation is zero, so any subsequent entry contributes to R_i .

Applying proportional allocation to total sample imports gives $B_i = S_{i,a}(-\Delta S_C)/(1 - S_{C,a})$ for rivals and $B_C = \Delta S_C$. We can then express $P_i = B_i + (P_i - B_i)$. The difference $P_i - B_i$ arises because an exporter's product mix affects its exposure to China's share changes. It depends on which products the exporter sells, China's change in each product, and the final product weights. Table A7 reports both benchmarks. We measure relative allocation within products by $(\Delta S_i - C_i)/P_i$, which equals one under proportional allocation.

Pakistan's within-product relative allocation moves from -4.35 in 2015–2017 to 2.20 in 2018–2019. Honduras moves from -1.04 to 1.61 and Cambodia from -2.53 to 2.05 . Mexico moves in the opposite direction, from 3.69 to -3.28 . Bangladesh's ratio rises from 0.12 to 1.78 . Its pre-tariff aggregate share loss comes from product spending weights, while its within-product change is slightly positive. Vietnam receives more than its proportional allocation in both periods, but its ratio falls from 5.15 to 1.92 .

Half the sum of absolute allocation residuals is 1.922 percentage points in 2015–2017 and 0.958 in 2018–2019. Dividing by the number of years gives 0.961 and 0.958 points a year. These amounts equal 108.1 and 32.4 percent of China's within-product decline. By this measure, the tariffs coincide with a change in which suppliers gain, without an increase in the annual amount of non-proportional reallocation.

Table A7. Within-Product Allocation before and during the Tariffs

<i>Panel A. Total change and within-product allocation</i>								
Exporter	2015–2017				2018–2019			
	ΔS_i	C_i	$\Delta S_i - C_i$	Ratio	ΔS_i	C_i	$\Delta S_i - C_i$	Ratio
China	-1.577	0.202	-1.779	—	-2.702	0.257	-2.959	—
Vietnam	1.401	-0.132	1.533	5.15	0.945	0.017	0.928	1.92
India	0.606	0.018	0.589	2.63	0.302	-0.212	0.514	1.23
Bangladesh	-0.152	-0.166	0.014	0.12	0.207	-0.033	0.240	1.78
Mexico	0.549	0.118	0.431	3.69	-0.350	-0.024	-0.326	-3.28
Indonesia	-0.217	-0.107	-0.110	-0.60	-0.072	-0.053	-0.019	-0.09
Pakistan	-0.171	-0.018	-0.154	-4.35	0.175	-0.008	0.183	2.20
Cambodia	-0.207	-0.010	-0.197	-2.53	0.309	0.005	0.304	2.05
Honduras	-0.124	-0.026	-0.097	-1.04	0.231	0.081	0.151	1.61
Italy	-0.047	-0.109	0.062	1.14	0.017	-0.034	0.051	0.40
Rest of world	-0.062	0.231	-0.293	-0.50	0.938	0.006	0.933	0.80

<i>Panel B. Proportional benchmarks and residuals</i>								
Exporter	2015–2017				2018–2019			
	B_i	$P_i - B_i$	P_i	R_i	B_i	$P_i - B_i$	P_i	R_i
China	-1.577	-0.202	-1.779	0.000	-2.702	-0.257	-2.959	0.000
Vietnam	0.219	0.078	0.298	1.235	0.430	0.054	0.483	0.444
India	0.190	0.034	0.224	0.365	0.340	0.078	0.419	0.096
Bangladesh	0.078	0.037	0.115	-0.101	0.124	0.011	0.135	0.105
Mexico	0.088	0.029	0.117	0.314	0.159	-0.060	0.099	-0.425
Indonesia	0.114	0.070	0.184	-0.293	0.170	0.039	0.209	-0.228
Pakistan	0.069	-0.034	0.035	-0.189	0.109	-0.026	0.083	0.100
Cambodia	0.057	0.022	0.078	-0.275	0.093	0.055	0.148	0.156
Honduras	0.070	0.023	0.093	-0.191	0.113	-0.019	0.093	0.057
Italy	0.048	0.007	0.055	0.008	0.082	0.046	0.128	-0.077
Rest of world	0.643	-0.063	0.581	-0.873	1.083	0.078	1.161	-0.229

Notes. Both periods use 1,361 products and all origins. Entries are percentage points except the relative allocation. The identities are $\Delta S_i = C_i + P_i + R_i$ and $P_i = B_i + (P_i - B_i)$.

D.4 Exposure Estimates and Supplier Exclusions

Table A8 reports exposure coefficients relative to Mexico and two checks that exclude suppliers. The first tests differences among the remaining coefficients in the original regression. The second removes the supplier's observations and re-estimates the regression. These joint tests use the estimation sample, whereas the prediction exercise in the main text uses separate estimation and evaluation samples.

Table A8. Unrestricted Exposure Responses and Supplier Sensitivity

Exporter	Slope relative to Mexico (SE)			2021 sample		
	2015–2017	2017–2019	2017–2021	Pairs	Mean X_p	SD X_p
Vietnam	-0.65 (4.27)	8.50 (3.38)	15.62 (3.55)	726	0.0376	0.0359
India	1.13 (2.39)	-0.59 (2.78)	0.46 (2.85)	1126	0.0403	0.0393
Bangladesh	0.23 (4.10)	5.21 (3.44)	5.74 (4.96)	454	0.0299	0.0296
Mexico	—	—	—	893	0.0385	0.0376
Indonesia	2.26 (3.57)	5.83 (4.27)	9.98 (4.01)	704	0.0370	0.0354
Pakistan	0.48 (3.70)	2.34 (3.54)	2.57 (5.14)	757	0.0387	0.0382
Cambodia	1.60 (4.09)	2.23 (4.48)	7.29 (7.18)	344	0.0302	0.0261
Honduras	-8.79 (8.24)	-2.06 (7.17)	-4.27 (6.16)	229	0.0370	0.0366
Italy	-0.34 (2.58)	0.10 (2.52)	-6.09 (2.75)	1177	0.0400	0.0393
Joint F	1.05	1.67	7.29			
Joint p	0.399	0.111	4.1×10^{-8}			
Pairs	6,133	6,231	6,410			

2021 equal-response test, p -value		
Excluded supplier	Full-model contrasts	Delete and refit
Vietnam	7.3×10^{-5}	1.8×10^{-4}
India	8.0×10^{-8}	9.5×10^{-7}
Bangladesh	1.9×10^{-8}	1.6×10^{-8}
Mexico	2.1×10^{-8}	3.3×10^{-7}
Indonesia	2.0×10^{-8}	1.2×10^{-7}
Pakistan	2.1×10^{-8}	2.0×10^{-7}
Cambodia	3.9×10^{-8}	2.0×10^{-7}
Honduras	8.0×10^{-7}	7.6×10^{-6}
Italy	4.7×10^{-5}	7.0×10^{-5}
Vietnam + Italy	0.129	0.191

Notes. All PPML estimates use 2021 exposure. Standard errors cluster at the four-digit heading level. The lower panel compares contrast tests in the full model with regressions re-estimated after supplier removal.

D.5 Changes in Exposure Responses

We account for correlation between estimates from overlapping periods when testing whether the exposure response changes. Let u_{wg} be the CR1-adjusted cluster influence vector for period w and four-digit heading g . We calculate it from the profiled PPML information matrix and cluster scores, with the same adjustment for nuisance effects used in the original covariance estimator. We verify that $\sum_g u_{wg} u_{wg}^\top$ matches each saved covariance matrix. The covariance of $\hat{\beta}_b - c\hat{\beta}_a$ is then

$$\hat{V}_{b-ca} = \sum_g (u_{bg} - cu_{ag})(u_{bg} - cu_{ag})^\top. \quad (\text{D.4})$$

We join the influence vectors by their four-digit product codes and assign zero scores to codes absent from a period. The tests use 144 degrees of freedom. Setting $c = 1$ compares cumulative coefficients, while setting $c = 2$ compares the four-year tariff-period response with twice the two-year pre-tariff response. The latter asks whether the earlier exposure trend continues at

the same annual rate, without imposing that trend in estimation. An additional audit refits the regressions after shuffling observations and separately scrambles the cluster rows before joining by code. It leaves the standard errors unchanged. The influence correlations with the 2015–2017 Vietnam–Mexico contrast are -0.077 for 2017–2021 and -0.347 for 2017–2019. Shared initial observations do not require these correlations to be equal, because the final outcomes, informative pairs, and PPML weights differ.

Table A9. Changes in the Vietnam–Mexico Exposure Contrast

Contrast	Difference	Standard error	<i>p</i> -value
2021 minus pre-tariff	16.27	5.76	0.005
2019 minus pre-tariff	9.15	6.30	0.149
2021 minus twice pre-tariff	16.91	9.49	0.077

Notes. The covariance calculation accounts for observations that appear in more than one period. The last row compares annual rates by extrapolating the two-year pre-tariff slope.

We also replace China’s 2017 share in the exposure measure with its 2015 share. We calculate this share from all origins in the original 2015 ten-digit files and aggregate to the same eight-digit codes. We keep tariff changes, product selection, fixed effects, and clustering unchanged. Vietnam’s coefficients relative to Mexico are -0.54 (4.56), 10.41 (3.38), and 16.75 (3.64) for the pre-tariff, early-tariff, and tariff-period periods. This separates the base year of exposure from the base year of the outcome. The 2015 shares can still reflect earlier economic conditions.

To express the coefficient as an import response, we use the standard deviation of 2021 exposure among informative Vietnam-product pairs, 0.035912 . Exponentiating the contrast times this standard deviation multiplies Vietnam’s expected final-to-initial import ratio relative to Mexico’s by 1.7524 . Applying the same transformation to the t_{144} confidence interval gives 1.3621 – 2.2545 . Thus the 75.24-percent increase describes the ratio of growth factors, rather than a difference in percentage growth rates.

D.6 Trade Elasticities under Alternative Demand Systems

We illustrate the demand implications of the exposure coefficients by assuming the same pass-through of Chinese tariff changes across products, holding rival prices fixed, and interpreting the coefficients as local demand derivatives. At full pass-through, we set $a_M = -\hat{\beta}_{\text{Italy}} + 0.1$ and $a_i = a_M + \hat{\beta}_i$ for the nine named rivals, with Mexico’s exposure coefficient normalized to zero. We fix $a_C = 1.5$ and $a_{\text{Other}} = a_M$. These choices match the estimated parameter differences among rivals and make every a_i positive. We combine these product-level estimates with the exact 2018 shares from the aggregate USITC table, which uses a different aggregation.

Given the parameters a , we calculate trade elasticities from main-text equation (15). China's share of Vietnam's marginal expenditure-share loss is

$$\mathcal{D}_{C \leftarrow V} = \frac{s_C}{1 - s_V} \frac{a_C + a_V - \sum_i s_i a_i}{\varepsilon_V}. \quad (\text{D.5})$$

We calculate standard errors by the delta method using the full covariance matrix of the eight exposure coefficients. These standard errors condition on the shares, pass-through, and the parameter levels chosen above. They do not account for uncertainty about those choices. The local perturbations used in the delta method do not impose inequality constraints on the estimates.

We check sensitivity using the two corrected estimates of China's direct response, $a_C = 0.7245$ and 2.1212 . We set Italy's parameter to 10^{-6} or three and the residual group's parameter to Mexico's or the share-weighted mean of the nine rivals. Across these eight cases, Vietnam's trade elasticity is 19.97–23.07 and its ratio to China's is 3.36–4.96. China receives 28.0–34.7 percent of Vietnam's marginal share loss. These ranges summarize sensitivity to the selected parameter choices.

For nested CES, let $S = s_C + s_V$, $v = s_V/S$, and let the between- and within-nest elasticities be $1 + \eta$ and $1 + \omega$. We match the local Vietnam–Mexico contrast at this benchmark by setting $\omega = \eta + S\hat{\beta}_V$. A Vietnam-only price change gives

$$-\frac{\partial \log s_V}{\partial \log p_V} = \omega(1 - v) + \eta(1 - S)v, \quad (\text{D.6})$$

$$\frac{\partial \log s_C}{\partial \log p_V} = v[\omega - \eta(1 - S)]. \quad (\text{D.7})$$

Taking the ratio of these responses and multiplying by s_C/s_V gives China's diversion share. Substituting $\omega = \eta + S\hat{\beta}_V$ gives the closed forms in main-text equation (23). We use $\eta = 1.5$ and three, which imply within-nest elasticities of 9.85 and 11.35. With fixed nest elasticities, the exposure coefficient is $(\omega - \eta)/S_p$ and therefore differs across products with different S_p . The exercise provides an alternative local demand interpretation at the aggregate benchmark.

The local approximation can understate the parameter difference needed for a finite response in the maintained single-price model. Let $f(x) = \log \hat{P}$ when only China's log price increases by x . Since $f'(x) = s_C(x)$ and $f''(x) = -s_C(x)[1 - s_C(x)]\varepsilon_C(x) < 0$, we have $0 < f(x) < s_C(0)x$ for $x > 0$. The exact relative-share change is $(a_V - a_M)f(x)$, so dividing it by $s_C(0)x$ gives less than $a_V - a_M$. With the normalized illustration and gross price changes 1.225/1.15 and 1.40/1.15, the exact index change is 91.3 and 75.8 percent of this first-order approximation. This argument applies to the specified price experiment. It does not establish a bound for a regression that combines products with other changes in demand, prices, or supply.

D.7 Simultaneous Price Changes

The later U.S. tariff episodes change several suppliers' prices and cannot be treated as an isolated increase in Vietnam's price. Effective August 7, 2025, the announced additional rates were 20 percent for Vietnam and Bangladesh and 19 percent for Cambodia, Indonesia, and Pakistan. Following the February 20, 2026 Supreme Court decision on IEEPA tariffs, the administration announced a temporary 10-percent surcharge. The Section 301 action effective July 24, 2026 set additional rates of 12.5 percent for Vietnam and China and 10 percent for Bangladesh, Cambodia, India, Indonesia, Pakistan, and Mexico, subject to product and other exemptions ([The White House, 2025](#), [Office of the United States Trade Representative, 2026a,b](#)).

We calculate the relative-share response to any vector $x_i = \Delta \log p_i$ using the same benchmark and parameters as the own-price illustration. Let $z = \Delta \log P$. Our share equations give

$$\Delta \log(s_V/s_j) = -a_V x_V + a_j x_j + (a_V - a_j)z. \quad (\text{D.8})$$

For the nest, its price-index change and Vietnam's relative share against any outside supplier satisfy

$$z_N = -\frac{1}{\omega} \log \left[\frac{s_C}{S} e^{-\omega x_C} + \frac{s_V}{S} e^{-\omega x_V} \right], \quad (\text{D.9})$$

$$\Delta \log(s_V/s_j) = -\omega x_V + \eta x_j + (\omega - \eta)z_N. \quad (\text{D.10})$$

Setting $x_V = x_j$ gives main-text equations (24)–(25). Other suppliers' prices enter the relative response in our model through z . In the nested model, the top-level index cancels from the relative-share equation, leaving the China–Vietnam index z_N .

Table A10 specifies three hypothetical tariff vectors and their exact relative-share predictions. We use initial gross duties of 1.30 for China and 1.15 for every other supplier, full pass-through, and the 2018 shares. If t_i denotes the additional tariff in the table, the log price change is $x_i = \log[(1 + \tau_i^0 + t_i)/(1 + \tau_i^0)]$. Each column is a separate counterfactual from this benchmark, not a transition from the preceding column. The residual group has an assumed rate, and Mexico is assumed exempt from the added duty. These choices follow the supplied numerical illustration and are not a reconstruction of actual tariff schedules. In particular, the statutory July 2026 rate for Mexico is 10 percent on covered goods, with exemptions determined by the policy rather than by the blanket assumption in the illustration.

Across the six Vietnam–rival responses, our model gives changes ranging from -0.56 to -0.15 log points, compared with -0.13 to $+0.26$ for the nests. Their signs and magnitudes depend on the entire specified price vector. The table shows how the models give different predictions when several prices change. Applying these predictions to a policy episode requires observed product-level duties, exemptions, and pass-through.

Table A10. Relative Import Shares under Simultaneous Price Changes

Supplier or outcome	A	B	C
<i>A. Additional tariff rates (percentage points)</i>			
China	30	20	12.5
Vietnam	20	20	12.5
India	25	25	10
Bangladesh	20	20	10
Mexico	0	0	0
Indonesia	19	19	10
Pakistan	19	19	10
Cambodia	19	19	10
Honduras	10	10	10
Italy	3	3	5
Other origins	12	12	11
<i>B. Change in Vietnam's relative share (log points)</i>			
Bangladesh, HCDE	-0.154	-0.374	-0.425
Bangladesh, Nest, $\eta = 1.5$	+0.259	-0.101	-0.096
Bangladesh, Nest, $\eta = 3$	+0.257	-0.101	-0.126
Pakistan, HCDE	-0.269	-0.560	-0.422
Pakistan, Nest, $\eta = 1.5$	+0.248	-0.112	-0.096
Pakistan, Nest, $\eta = 3$	+0.234	-0.123	-0.126

Notes. Hypothetical vectors, not measured policy regimes. Initial gross duties are 1.30 for China and 1.15 for every other group. Each scenario begins at the same benchmark. The calculations assume full pass-through, an exemption from the added duty for Mexico, and the stated residual-group rates. Our model uses the normalized product-based illustration. Each nest matches the Vietnam–Mexico exposure contrast.

D.8 Tests of the Aggregate Calibration

The log-change counterpart to PPML is

$$\Delta \log s_{ip} = \alpha_p + \gamma_i + \kappa d_i X_p + u_{ip}. \quad (\text{D.11})$$

We cluster standard errors at the four-digit heading level. Table A11 reports estimates that restrict the pattern of exposure responses across exporters to the aggregate calibration. We test whether the coefficient on this pattern equals zero or 0.77. The ratio $\hat{\kappa}/0.77$ compares the estimated magnitude with the calibration's prediction. It does not measure the share of variation explained.

Table 1 reports unrestricted PPML estimates with HTS 63079098 excluded. All three columns use $X_{p,2021}$, so the coefficients have the same units. If we use the tariff exposure for each period, the joint tests give $p = 0.088$ in 2019 and $p = 0.311$ in the placebo. Conditional on its estimated covariance, the earlier 2019 OLS test of eight restrictions has about 22 percent power against the calibrated pattern, or 10.7 percent after adjustment for trends.

Table A11. Exposure Responses Predicted by the Aggregate Calibration

Specification	Pairs	$\hat{\kappa}$ (SE)	$\hat{\kappa}/0.77$	p -value		
				$\kappa = 0$	$\kappa = 0.77$	Equal γ_i
PPML, 2021, without 63079098	6,410	0.999 (0.283)	1.30	< 0.001	0.419	< 0.001
PPML, 2020, without 63079098	6,279	1.102 (0.296)	1.43	< 0.001	0.262	< 0.001
PPML, 2017–2019	6,285	0.623 (0.321)	0.81	0.055	0.647	0.032
OLS, 2017–2019	4,224	0.775 (0.436)	1.01	0.077	0.990	0.580
Value-weighted, 2017–2019	4,224	0.277 (0.402)	0.36	0.493	0.223	0.010
Pooled ROW, 2017–2019	5,583	0.767 (0.403)	1.00	0.059	0.994	0.192
Value-weighted, 2017–2021	4,202	0.751 (0.366)	0.98	0.042	0.960	< 0.001
OLS, 2015–2017 placebo	4,224	-0.146 (0.338)	-0.19	0.666	0.008	0.319
PPML, 2015–2017 placebo	6,195	-0.154 (0.438)	-0.20	0.725	0.036	< 0.001

Notes. All rows use the fixed aggregate direction. Standard errors cluster by heading. Weighted rows use 2017 imports. The ratio compares coefficient scales, not explained variation.

Let b collect the eight 2021 slopes and V their covariance. The covariance-weighted projection is $\tilde{\kappa} = (d'V^{-1}b)/(d'V^{-1}d)$, giving

$$b'V^{-1}b = \tilde{\kappa}^2 d'V^{-1}d + (b - \tilde{\kappa}d)'V^{-1}(b - \tilde{\kappa}d).$$

We obtain $\tilde{\kappa} = 0.69$ (0.21). This differs from the restricted PPML coefficient because the two estimators use different weights. The residual test imposes seven restrictions, holding d fixed. We report both tests on subsets of the full-sample coefficient vector and tests after removing suppliers' observations and re-estimating.

The restricted PPML coefficient in 2021 is 1.00 (0.28), with $p = 0.419$ against 0.77. Figure A5 shows the annual estimates. These results concern the magnitude of a response pattern imposed by the calibration. The unrestricted coefficients still reject proportional allocation.

We also estimate a separate response for each tariff list using $d_{iSCp,2017} \sum_{\ell} b_{\ell} \mathbf{1}\{p \in \ell\}$, with the same exporter effects across lists. Table A12 reports coefficients in the same units. The List 3 and List 4A coefficients differ at $p = 0.016$, accounting for their covariance. A test that responses are proportional to scheduled tariffs gives $p = 0.589$ when the coefficient is estimated freely and $p = 0.786$ when it is fixed at the aggregate calibration's prediction.

List 4B is announced for December 15, 2019 and suspended before implementation, so anticipation can affect products whose additional duty remains zero (Office of the United States Trade Representative, 2019a,b). Adding a separate $d_{iSCp,2017}$ term gives a tariff coefficient of 0.653 (0.432) and an initial-China-share coefficient of -0.003 (0.027). This specification separates tariff exposure from initial Chinese specialization, but estimates the tariff response less precisely.

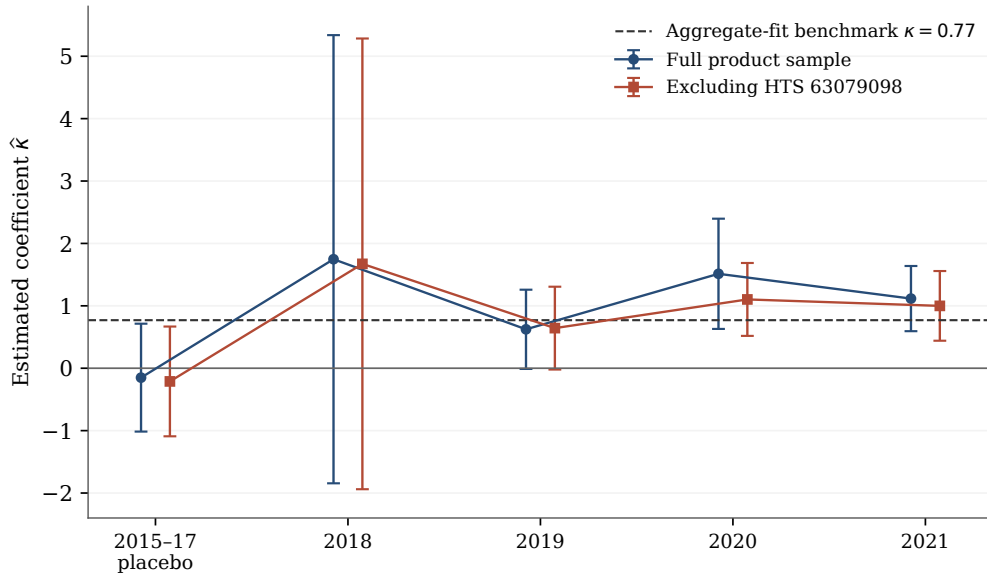


Figure A5. Estimated Exposure Responses over Time

Notes. PPML coefficients with pointwise 95-percent confidence intervals clustered by four-digit heading. Each tariff-year estimate uses that year's annual-average exposure. The placebo uses 2019 exposure.

Table A12. Exposure Responses by Tariff List

Policy bin	Coefficient (SE)		PPML model projection
	PPML	OLS	
List 3	0.148 (0.053)	0.129 (0.075)	0.132
List 4A	0.019 (0.021)	0.010 (0.049)	0.033
List 4B	0.028 (0.047)	0.018 (0.040)	0.001
Equal List 3 and List 4A	0.016	0.085	—
All bins equal	0.054	0.225	—
Proportional to scheduled dose	0.589	0.636	—
Aggregate-fit coefficient	0.786	0.820	—

Notes. Coefficients multiply $d_{iSCp,2017}$, with the same exporter effects across tariff lists. The lower rows report Wald tests using the estimated covariance matrix. Inference conditions on the calibrated response pattern and projection weights.

D.9 China's Own Response

For customs-value shares, the ten-supplier regression is

$$\Delta \log s_{ip}^V = \alpha_p + \gamma_i + \beta_i X_p + c_V \mathbf{1}\{i = C\} \Delta \ell_{p,2019} + u_{ip}, \quad \beta_M = 0. \quad (\text{D.12})$$

Duty-inclusive expenditure is $E_{ip} = (1 + \tau_{ip}) V_{ip}$. Growth in a source's customs-value share therefore equals growth in its expenditure share minus its log tariff change, plus a term that is the same for all sources of the product. We apply the local approximation from our model, assume constant pass-through ϕ of Chinese tariff changes across products and unchanged rival prices and tariffs, and treat the scheduled tariff change as the change in duties actually paid. These assumptions give

$$c_V = -\phi a_C - 1, \quad \beta_i = \phi(a_i - a_M), \quad \phi a_i = \beta_i - \beta_C - c_V - 1. \quad (\text{D.13})$$

Adding China's scheduled tariff change to the outcome raises c_V by one, leaving other coefficients and their covariance unchanged.

Under these additional restrictions, equation (D.13) determines parameter levels at a chosen pass-through rate. Some point estimates violate positivity. At full pass-through, the unweighted 2017–2019 regression implies $a_{\text{Cambodia}} = -10.90$. The value-weighted regression implies $a_{\text{Cambodia}} = -0.22$, $a_{\text{Honduras}} = -0.35$, and $a_{\text{Italy}} = -1.30$. These are unrestricted point estimates, not inequality tests, but they show that the larger regression does not automatically yield a parameter vector in the maintained demand family. The normalized illustration in Section D.6 uses the nine-rival 2021 contrasts and explicitly chosen levels instead of imposing this ten-supplier fit.

Table A13. China's Direct Scheduled-Tariff Response

Specification	c_V	SE	$-c_V - 1$
Unweighted, 2017–2019	-1.725	0.812	0.725
Value-weighted, 2017–2019	-3.121	0.890	2.121
Unweighted, net of growth	-0.536	1.364	-0.464
Value-weighted, net of growth	-3.629	1.558	2.629

Notes. Ten suppliers enter (D.12). Interpreting $-c_V - 1$ as $\phi(\sigma_C - 1)$ requires scheduled changes to represent the actual duty wedge.

D.10 Out-of-Sample Prediction

A response pattern estimated from import growth gives a median coefficient of 0.02 across twenty sample splits. Import growth without price variation does not separate substitution parameters from supplier trends.

We use the same 1,213 products throughout and identify the 25 largest non-China origins by 2017 imports, pooling the remainder. We make 299 random splits into groups of 72 and 73 four-digit headings. Each named origin must have positive imports in 2015 and at least \$10 million in 2017 in both halves. We estimate the placebo before the tariff regressions. In each estimation sample, we normalize Mexico’s coefficient to zero and estimate a separate coefficient for the residual group. We then use the coefficients from one half to predict responses in the other through $\lambda_f \hat{\beta}_i^{(-f)} X_{p,2021}$, with separate product and exporter effects. We give the two estimated prediction coefficients equal weight within each split. Estimation error in the first sample can attenuate the coefficient in the second.

We combine the 598 one-sided cluster- t tests from the evaluation samples. Following the fixed half-quantile rule in [Meinshausen, Meier and Bühlmann \(2009\)](#), we use twice the upper median of their p -values, capped at one. Rejection at α requires at least half the tests to have $p_f \leq \alpha/2$. Markov’s inequality then controls test size if each conditional half-sample test is valid, even though the splits overlap. The PPML tests use CR1 standard errors clustered by four-digit heading and $G_f - 1$ degrees of freedom. Their validity depends on this approximation and independence across headings. We use the authors’ rule for combining tests, without applying their high-dimensional variable-selection procedure.

Main-text Table 2 repeats the exercise with supplier exclusions and for the earlier period. We keep the sample splits, minimum import requirement, residual group, and exposure measure unchanged. Excluding a supplier removes its observations from both estimation and evaluation samples, without adding smaller origins. The 2019 row uses 2021 exposure, as in Table 1. All rows use the same 299 splits. The distribution across splits shows how the result depends on the division of products.

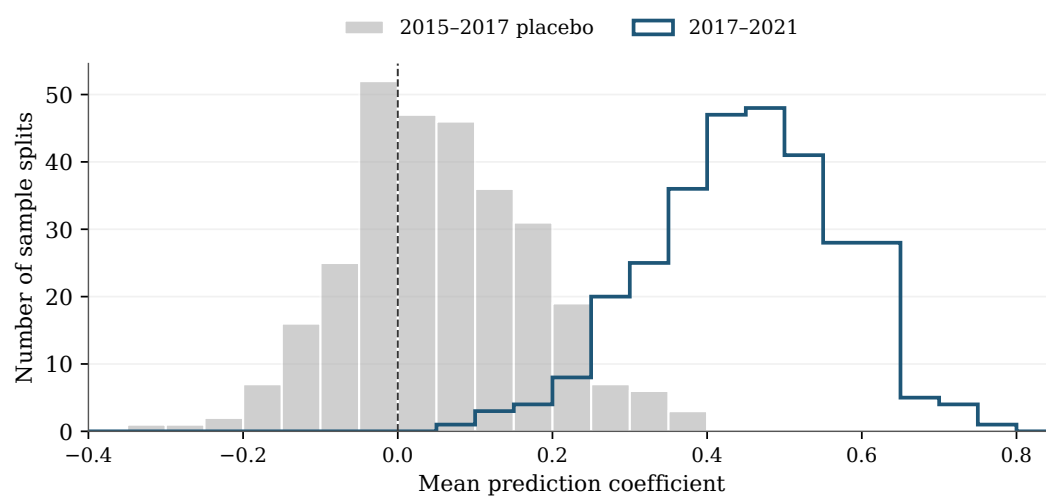


Figure A6. Out-of-Sample Predictions across Random Product Splits

Notes. Each period uses the same 299 splits of four-digit headings and 2021 exposure. Histograms show sensitivity to the division of products.

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